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Overview

This course is sort of a sort of introductory but not completely introductory course in Linear algebra; perhaps a second course. Knowledge of my the vectors and matrices sublevel and matrix topics in earlier sublevels will be assumed, particularly in chapter 4 where most of the content can be found in level 6.1 for the determinant section and Cayley-Hamilton theorem proof and 7.3 for the endomorphism section, specifically in the Jordan normal form proof.

1 Vector spaces

1.1 Definitions

Definition. A field $\mathbb{F}$ is a set with binary operations $+$ and $*$ satisfying

1. $+$ and $*$ are both associative and commutative
2. $+$ and $*$ have identity elements 0 and 1 respectively
3. Every element has a $+$ -inverse and every non-zero element has a $*$ -inverse
4. Multiplication distributes over addition so that

$$r_1 * (r_2 + r_3) = (r_1 * r_2) + (r_1 * r_3)$$

$$(r_1 + r_2) * r_3 = (r_1 * r_3) + (r_2 * r_3)$$

Example. $\mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ are fields. If p is prime, then the integers mod p form a field.

The way to think about a field is as a number system, as opposed to a group which is about symmetries.

Definition. A vector space over a field $\mathbb{F}$ is a set of vectors that is closed under addition and closed under multiplication by scalars which are elements of $\mathbb{F}$. A vector space V over $\mathbb{F}$ with $v, u \in V$ and $\lambda, \mu \in \mathbb{F}$ satisfies

1. $\lambda(\mu v) = (\lambda\mu)v$
2. $\lambda(u + v) = \lambda u + \lambda v$
3. $(\lambda + \mu)v = \lambda v + \mu v$
4. $1v = v$

In each of the following examples the operations are as you expect

Example. $\mathbb{R}^n$ is a vector space for $n \in \mathbb{N}$

Example. For a set X the set of functions $f : X \rightarrow \mathbb{R}$ is a vector space.

Example. Define $C([a, b])$ as the set of continuous functions $[a, b] \rightarrow \mathbb{R}$, then this is a vector space.

Example. The set of $m \times n$ matrices with coefficients in $\mathbb{R}$ is a vector space.

Note that there is not always a notion of coordinates, length, angle or distance. It is also easy to check that $0v = 0$ for all v where 0 is the additive identity in the vector space, and $(-1)v = -v$.

Definition. If V is a vector space over a field $\mathbb{F}$ then $U \subseteq V$ is a vector subspace if it is non-empty and remains closed under addition and scalar multiplication.

Example. $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ is a subspace of $\mathbb{R}^3$.

Example. Let X be any set. The set of functions $X \rightarrow \mathbb{R}$ such that finitely many elements of X map to a non-zero element of $\mathbb{R}$ is a subspace of the set of functions $X \rightarrow \mathbb{R}$. We say these are the functions with finite support.

Note that the intersection of subspaces is a subspace but the union is not necessarily a subspace (consider 2 planes through the origin that's clearly not a subspace).

Definition. Let U, W be subspaces of a vector space V , then define their sum as

$$U + W = \{u + w : u \in U, w \in W\}$$

Note that this is a subspace because to check if S is a subspace we need to check that if we have $u, v \in S$ and $\lambda, \mu \in \mathbb{F}$ then $\lambda u + \mu v \in S$. Indeed this works for $U + W$ because

$$\lambda(u_1 + w_1) + \mu(u_2 + w_2) = (\lambda u_1 + \mu u_2) + (\lambda w_1 + \mu w_2) \in U + W$$

because U, W are subspaces. It is non-empty as it contains 0 .

Definition. If V is a vector space and U is a subspace then the subgroup V/U is the quotient space, where scalar multiplication is given by $(\lambda, v + U) = (\lambda v) + U$.

This is well defined since if $v + U = w + U \in V/U$ then $v - w \in U$ so for $\lambda \in \mathbb{F}$ we have $\lambda v - \lambda w \in U$ so $\lambda v + U = \lambda w + U$.

1.2 Linear independence and basis

In the Groups Rings and Modules course you will see that in a certain generalization of Vector spaces dimension doesn't play as nice as it does here.

Definition. Let V be a vector space and S be a subset, then the span of S is the smallest subspace of V containing S or equivalently the set of finite linear combinations of elements of S . We write this as $\text{span}(S)$ or $\langle S \rangle$.

Note that we won't deal with infinite sums since we have not defined convergence in a general vector space.

Example. If $V = \mathbb{R}^3$ and $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$ then

$$\langle S \rangle = \text{span}(S) = \left\{ \begin{pmatrix} a \\ b \\ b \end{pmatrix} : a, b \in \mathbb{R} \right\}$$

Example. Let X be a set and $x \in X$. Define $\delta x : X \rightarrow \mathbb{F}$ as $\delta x(y) = 1$ if $y = x$ and 0 otherwise. Then the span of $\delta x : x \in X$ is the set of all functions with finite support.

Definition. S is a spanning set of V or S spans V if $\text{span}(S) = V$.

Definition. If S is a subset of V such that $\sum \lambda_i s_i = 0 \implies \forall i (\lambda_i = 0)$ then S is linearly independent.

Definition. A basis for V is a linearly independent spanning set.

Definition. A vector space is finite dimensional if there is a finite basis. Its dimension is the number of vectors in a basis but we ought to prove this is well defined.

Example. If $V = \mathbb{R}^3$ and $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$ then S is linearly independent because

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = 0$$

S also does not span V because $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is missing.

Note that no linearly independent set can contain 0 as $1 * 0 = 0$ and $\text{span}(\emptyset) = \{0\}$.

Note that a set S is linearly dependent if and only if there exist $s_0, s_1, \dots, s_n \in S$ and $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ such that

$$\sum_{i=1}^n \lambda_i s_i = s_0$$

Note that n may be 0 and this characterization will still hold as the statement becomes equivalent to $s_0 = 0$. In general it holds because if S is linearly dependent then there is a non-empty finite subset of s where a non-zero linear combination is 0 and we can call one s_0 and divide¹ by its coefficient.

Proposition. If S is a subset of a vector space V then it is a basis if and only if every $v \in V$ can be written uniquely as a finite linear combination of elements of S .

Proof. If S is a basis then it is a spanning set so every v can be written as a linear combination of elements of S . In particular there exists $e_i \in S$ and $n \in \mathbb{N}$ such that $v = \sum_{i=1}^n \lambda_i e_i$. If this is not unique then there exist f_j such that $v = \sum_{i=1}^n \lambda_i e_i = \sum_{j=1}^m \mu_j f_j$. Re-indexing so that the e_i 's are all the e 's and f 's and renaming the scalars accordingly we have

$$v = \sum_{i=1}^N \lambda_i e_i = \sum_{i=1}^N \mu_i e_i$$

This implies

$$\sum_{i=1}^N (\lambda_i - \mu_i) e_i = 0$$

so that all $\lambda_i = \mu_i$ by linear independence establishing uniqueness.

¹By the definition of a field you can "divide" by anything that isn't 0

On the other hand if everything is a finite linear combination in a unique way then we have a basis by essentially the reverse logic - if S was not spanning everything would not be a finite linear combination and if S was not linearly independent we could contradict uniqueness by adding a sum of finitely many elements of S that goes to 0.

□

Theorem. (Steinitz exchange lemma) Let V be a vector space over $\mathbb{F}$ and let $S = \{e_1, e_2, \dots, e_n\}$ be a finite linearly independent subset and T a spanning subset. Then there is some $T' \subseteq T$ of size n such that $(T \setminus T') \cup S$ still spans V . In particular, $|T| \geq n$,

Proof. Copy the proof of the following lemma from my vectors and matrices notes

“If $\{v_1, v_2, v_3, \dots, v_n\}$ is linearly independent in a vector space V and $\{w_1, w_2, w_3, \dots, w_m\}$ spans V , then $n \leq m$ ”

since it is identical.

□

Corollary. If $\{e_1, e_2, \dots, e_n\}$ is a linearly independent subset of V and $\{f_1, f_2, \dots, f_m\}$ spans V then $m \geq n$ there is a re-ordering of the f_i such that $\{e_1, e_2, \dots, e_n, f_{n+1}, \dots, f_m\}$ spans V .

We immediately get that if V has a finite basis then every basis has the same size because each one must have size $\geq$ the other one.

Proposition. Let V be a vector space over $\mathbb{F}$ with a basis of size n . Then every linearly independent set of size n is a basis.

Proof. If T is a linearly independent subset of size n that does not span v then there is some $v \in V \setminus \text{span}(T)$ and $T \cup \{v\}$ is independent (since if a finite linear combination adds to 0 then if the coefficient of v is 0 this contradicts linear independence of T and otherwise this gives v as a linear combination of elements of T contradicting $v \in V \setminus \text{span}(T)$). This contradicts existence of a basis of size n

□

Proposition. Let V be a vector space over $\mathbb{F}$ with a basis of size n . Then every spanning set of size n is a basis.

Proof. If T is a spanning set of size n and it is linearly dependent then there is some finite linear combination that adds to 0 which implies one vector in T is a linear combination of the others and can be removed and we would still have a basis.

□

Proposition. Let V be a vector space over $\mathbb{F}$ with a basis of size n . Then every finite spanning set contains a basis.

Proof. If T is any finite spanning set and $T' \subseteq T$ is a spanning set of least possible size which exists because T is finite and $|T'|$ has size n then done by the previous proposition. Otherwise it has size $> n$ and must be linearly dependent which again means there is some finite linear combination that adds to 0 which implies one vector in T is a linear combination of the others and can be removed and we would still have a basis.

□

Proposition. Let V be a vector space over $\mathbb{F}$ with a basis of size n . Then every linearly independent subset of V can be extended to a basis.

Proof. If T is a linearly independent set then since S is a basis that spans there is some $S' \subseteq S$ of size $|T|$ such that $(S \setminus S') \cup T$ spans V by the Steinitz exchange lemma, and this gives a spanning set of size n extended from T which thus must be a basis.

□

Proposition. Let V be a finite dimensional vector space over a field $\mathbb{F}$ and $U \subseteq V$ a proper subspace. Then U is finite dimensional and $\dim(U) < \dim(V)$.

Proof. Every linearly independent subset of V has size at most $\dim(V)$ so let $S \subseteq U$ be a linearly independent subset of largest size. Then we hope to show that S spans U and $|S| < \dim(V)$.

To do this note that if $v \in V \setminus \text{span}(S)$ then $S \cup \{v\}$ is linearly independent so $v \notin U$ by maximality of S so indeed $\langle S \rangle = U$. Since $U \neq V$ there is some $v \in V \setminus U = V \setminus \langle S \rangle$ so $S \cup \{v\}$ is a linearly independent subset of order $|S| + 1$ which is therefore $\leq \dim(V)$ so that $|S| < \dim(V)$.

□

Ok now here's the first actually kind of somewhat interesting proposition.

Proposition. If U, W are subspaces of a finite dimensional vector space V then

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

Proof. Let $R = \{v_1, v_2, \dots, v_r\}$ be a basis for $U \cap W$. Then this is a linearly independent subset of U . Extend it to a basis of U by $S = \{v_1, \dots, v_r, u_{r+1}, \dots, u_s\}$ and obtain a basis for W by $T = \{v_1, \dots, v_r, w_{r+1}, \dots, w_t\}$. We want to show that $\dim(U + W) = s + t - r$ and to do this it is sufficient to prove that $S \cup T$ is a basis for $U + W$.

It is spanning because if for an arbitrary $u + w \in U + W$ with $u \in U$ and $w \in W$ it is a linear combination of elements of $S \cup T$.

To show linear independence suppose we have

$$\sum_{i=1}^r \lambda_i v_i + \sum_{j=r+1}^s \mu_j u_j + \sum_{k=r+1}^t \nu_k w_k = 0$$

then $\sum \lambda_i v_i + \sum \mu_j u_j = -\sum \nu_k w_k$. Since the left hand side is something in U and the right hand side is something in W they are both in $U \cap W$.

Since S is a basis of U there is only one way of writing $\sum \lambda_i v_i + \sum \mu_j u_j$ but since R is a basis of $U \cap W$ we can write this such that all μ_j 's are 0 so they all must be 0 so that we have $\sum_{i=1}^r \lambda_i v_i + \sum_{k=r+1}^t \nu_k w_k = 0$. Since T is linearly independent we must have that all the remaining coefficients are 0.

□

Proposition. If V is a finite dimensional vector space over a field $\mathbb{F}$ and $U \subseteq V$ is a subspace then we have

$$\dim(V) = \dim(U) + \dim(V/U)$$

Proof. Let $\{u_1, \dots, u_m\}$ be a basis for U and extend this to a basis $\{u_1, \dots, u_m, v_{m+1}, \dots, v_n\}$ for V . Then we want to show that $\{v_{m+1} + U, \dots, v_n + U\}$ is a basis for V/U .

For spanning: If $v + U \in V/U$ then we can write $v = \sum \lambda_i u_i + \sum \mu_j v_j$ so that

$$v + U = \sum \mu_j (v_j + U) + \sum \lambda_i (u_i + U) = \sum \mu_j (v_j + U)$$

For linear independence: If $\sum \lambda_i (v_i + U) = 0 + U$ then this implies that $\sum \lambda_i v_i \in U$ so that there are μ_j such that $\sum \lambda_i v_i = \sum \mu_j u_j$. Since $\{u_1, \dots, u_m, v_{m+1}, \dots, v_n\}$ is a basis for V we must have $\lambda_i = \mu_j = 0$ for all i, j so $\{v_i + U\}$ is linearly independent.

□

1.3 Direct sums

Definition. If V is a vector space over a field $\mathbb{F}$ and $U, W \subseteq V$ are subspaces then we say that V is the internal direct sum of U and W if $U + W = V$ and $U \cap W = 0$.

Equivalently this means that every $v \in V$ can be written uniquely as $u + w$ with $u \in U$ and $w \in W$ and we say that U and W are complementary subspaces of V .

Example. If $V = \mathbb{R}^2$ and $U = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle$ then $\left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle$ and $\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle$ are both complementary subspaces to U in V .

Definition. If U and W are vector spaces over $\mathbb{F}$ then their external direct sum is $U \oplus W := \{(u, w) : u \in U, w \in W\}$ with addition and scalar multiplication defined component-wise eg $(u_1, w_1) + (u_2, w_2) = (u_1 + u_2, w_1 + w_2)$.

Intuitively, the difference is that an internal direct sum happens in an ambient space and an external direct sum is creating a new space from old.

Note that the external direct sum is the internal direct sum of U and W viewed as subspaces of $U \oplus W$.

Definition. The multiple internal direct sum is defined as follows: If $U_1, U_2, \dots, U_n \subseteq V$ are subspaces of V then V is the internal direct sum if every $v \in V$ can be written uniquely as $v = \sum u_i$ with $u_i \in U_i$ i.e. we write

$$V = U_1 \oplus U_2 \oplus \dots \oplus U_n = \bigoplus_{i=1}^n U_i$$

Note that for finite dimensions the dimension of the direct sum is the sum of the dimensions. A multiple external direct sum is defined analogously, explicitly if $U_1, U_2, \dots, U_n$ are vector spaces over $\mathbb{F}$ their external direct sum is

$$U_1 \oplus U_2 \oplus \dots \oplus U_n = \bigoplus_{i=1}^n U_i = \{(u_1, u_2, \dots, u_n) : u_i \in U_i\}$$

with operations defined similarly to above.

This can be made into an infinite sum provided you require that all but finitely many of the u_i are 0, since sums have to be well defined and you don't want to be saying $(s_1, s_2, \dots) = (s_1, 0, \dots) + (0, s_2, \dots) + \dots$ as an infinite sum without a notion of convergence.

2 Linear maps and matrices

If U, V are vector spaces write $\mathcal{L}(U, V)$ for the set of linear maps $U \rightarrow V$.

Note that a linear map is a group homomorphism if we view vector spaces as groups under addition.

Example. Complex conjugation is a map $\mathbb{C} \rightarrow \mathbb{C}$ that is $\mathbb{R}$ -linear but not $\mathbb{C}$ -linear.

We know matrices are linear, but we can do the following more abstract examples.

Definition. If X is a set and $\mathbb{F}$ is a field let $\mathbb{F}^X$ be the set of functions $\mathbb{F} \rightarrow X$.

Example. Define $m_g : \mathbb{F}^X \rightarrow \mathbb{F}^X$ by $m_g(f)(x) = g(x)f(x)$. Then m_g is linear as for example $f(x) \mapsto xf(x)$ is linear.

Example. The map $I : C([a, b]) \rightarrow \mathbb{R}$ defined by $f \mapsto \int_a^b f(x) dx$ is linear.

Definition. If I is an interval let $C^k(I)$ be the set of k -times continuously differentiable functions on I and $C^\infty(I)$ be the intersection of all C^k . If an endpoint is attained consider the one-sided derivative.

Example. The map $I : C^\infty([a, b]) \rightarrow C^\infty([a, b])$ defined by $f \mapsto f'$ is linear.

Note that sums and scalar multiples of linear maps are linear so a set $\mathcal{L}(U, V)$ forms a vector space over $\mathbb{F}$.

Definition. We say a linear map $\alpha : U \rightarrow V$ is an isomorphism if it has a linear inverse

Proposition. if U and V are vector spaces over $\mathbb{F}$ and $\alpha : U \rightarrow V$ is a function then it is an isomorphism if and only if it is a bijective linear map.

Proof. If α is an isomorphism it is a bijection because it has an inverse function and it is linear by definition.

If α is a linear bijection then as a function it has an inverse $\beta : V \rightarrow U$ that we want to show is linear. Let $v_1, v_2 \in V$ and $\lambda, \mu \in \mathbb{F}$. Then we have

$$\alpha\beta(\lambda v_1 + \mu v_2) = \lambda v_1 + \mu v_2 = \lambda\alpha\beta(v_1) + \mu\alpha\beta(v_2) = \alpha(\lambda\beta(v_1) + \mu\beta(v_2))$$

Since α is injective we have

$$\beta(\lambda v_1 + \mu v_2) = \lambda\beta(v_1) + \mu\beta(v_2)$$

so β is linear. □

Definition. Let $\alpha : U \rightarrow V$ be a linear map. Then its image is its range and its kernel is the set of things that map to 0 and easy check is these are both subspaces.

Definition. $\text{Mat}_{m,n}(\mathbb{F})$ is the set of linear maps $\alpha : \mathbb{F}^n \rightarrow \mathbb{F}^m$ and we will shortly show that these are exactly the matrices.

Example. If $v \mapsto Av$ is a linear map α then the system of linear equations $\sum_{j=1}^m A_{ij}x_j = b_i$ has a solution if and only if $(b_1, b_2, \dots, b_n) \in \text{im}(\alpha)$ and the kernel of α is all solutions to $\sum_{j=1}^m A_{ij}x_j = 0$.

Example. Let $\beta : C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$ be defined as

$$\beta(f)(t) = f''(t) + p(t)f'(t) + q(t)f(t)$$

for some $p, q \in C^\infty(\mathbb{R})$.

Then if $y(t) \in \text{im}(\beta)$ then there is a solution in $C^\infty(\mathbb{R})$ to the differential equation $f''(t) + p(t)f'(t) + q(t)f(t) = y(t)$. Also, $\ker(\beta)$ contains the smooth solutions to the homogeneous equation $f''(t) + p(t)f'(t) + q(t)f(t) = 0$.

Proposition. Let $\alpha : U \rightarrow V$ be an $\mathbb{F}$ -linear map. Then if α is injective and $S \subseteq U$ is linearly independent then $\alpha(S)$ is linearly independent in V .

Proof. Suppose that α is injective and $\alpha(S)$ is linearly dependent in V so we want to show that S is linearly dependent in U . By definition there are $s_0, s_1, \dots, s_n \in S$ distinct and $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$ not all zero such that

$$\alpha(s_0) = \sum_{i=1}^n \lambda_i \alpha(s_i) = \alpha\left(\sum_{i=1}^n \lambda_i s_i\right)$$

□

Proposition. Let $\alpha : U \rightarrow V$ be an $\mathbb{F}$ -linear map. Then if α is surjective and $S \subseteq U$ spans U then $\alpha(S)$ spans V .

Proof. Exercise

□

Proposition. If α is an isomorphism and $S \subseteq U$ is a basis then $\alpha(S)$ is a basis for V .

Proof. Follows immediately from the previous 2 propositions

□

Corollary. If U and V are finite dimensional vector spaces over $\mathbb{F}$ and $\alpha : U \rightarrow V$ is an isomorphism then $\dim(U) = \dim(V)$.

Note also that there is a bijection between bases for isomorphic infinite dimensional vector spaces but we have not proved dimension is unique in this context.

Now we prove that any two finite-dimensional vector spaces are isomorphic.

Theorem. Suppose V is a $\mathbb{F}$ -vector space of dimension $n < \infty$. Then writing $e_1, e_2, \dots, e_n$ for the standard basis of $\mathbb{F}^n$ there is a bijection

$$\phi : \{\text{isomorphisms } \mathbb{F}^n \rightarrow V\} \rightarrow \{\text{ordered basis } (v_1, v_2, \dots, v_n) \text{ for } V\}$$

defined by

$$\alpha \mapsto (\alpha(e_1), \alpha(e_2), \dots, \alpha(e_n))$$

Proof. This is a function because previous work shows α sends a basis for a basis.

To prove injectivity:

Suppose $\alpha, \beta : \mathbb{F}^n \rightarrow V$ are isomorphisms such that $\phi(\alpha) = \phi(\beta)$. Then this implies that $\alpha(e_i) = \beta(e_i)$ for all i . We want to prove that this implies $\alpha = \beta$. We have

$$\alpha\left(\sum_{i=1}^n x_i e_i\right) = \sum x_i \alpha(e_i) = \sum x_i \beta(e_i) = \beta\left(\sum_{i=1}^n x_i e_i\right)$$

To prove surjectivity:

Suppose $(v_1, v_2, \dots, v_n)$ is an ordered basis for V then define $\alpha(\sum_{i=1}^n x_i e_i) = \sum_{i=1}^n x_i v_i$. Easy check is this is well defined and linear. We know α is injective because $(v_1, v_2, \dots, v_n)$ is linearly independent so if $\sum x_i v_i = \sum y_i v_i$ then $x_i = y_i$. Also, α is surjective because $(v_1, v_2, \dots, v_n)$ spans V so α is an isomorphism and by construction $\phi(\alpha) = (v_1, v_2, \dots, v_n)$

□

This is an interesting result - it basically says that all finite dimensional vector spaces with the same dimension over the same field are isomorphic. Geometrically, in $\mathbb{R}^n$ it is easy to see kind of why this should be the case, but this result extends to all fields.

Note that to define a linear map, it suffices to define its values on basis elements. To see that this extension from basis elements can be done and only in 1 way, if $S := \{e_i : i \in I\}$ is a basis for U and $\alpha : S \rightarrow V$ is a linear map where V is another vector space and α, β are linear maps $U \rightarrow V$ that agree on S , then a typical element of U is a finite sum of some e_i 's which is forced to have one value and only one value on V .

Because of this, if U and V are finite dimensional vector spaces over $\mathbb{F}$ with basis $(e_1, e_2, \dots, e_m)$ and $(f_1, f_2, \dots, f_n)$ respectively then there is a bijection $\text{Mat}_{n,m}(\mathbb{F}) \mapsto \mathcal{L}(U, V)$. In particular, the linear maps between finite vector spaces over the same field are exactly the matrices.

If a linear map is represented by a matrix A then the i 'th column of A tells us how to write $\alpha(e_i)$ in terms of the f_j by $\alpha(e_i) = \sum A_{ji}f_j$.

Now we start the arrowslop. Given a basis $e_1, e_2, \dots, e_m$ for a finite dimensional vector space we get a bijection $s(e_i) : U \rightarrow \mathbb{F}^m$ and given a basis f_i for $\mathbb{F}^n$ we get an isomorphism $s(f_i) : V \rightarrow \mathbb{F}^n$. Since a matrix is a linear map $A : \mathbb{F}^m \rightarrow \mathbb{F}^n$, given a matrix A we can make a linear map $\alpha : U \rightarrow V$ as shown in Figure 1:

$$U \xrightarrow{s(e_i)} \mathbb{F}^m \xrightarrow{A} \mathbb{F}^n \xrightarrow{s(f_i)^{-1}} V. \quad \text{Figure 1}$$

We can make this a square as shown in Figure 2:

$$\begin{array}{ccc} \mathbb{F}^m & \xrightarrow{A} & \mathbb{F}^n \\ s(e_i) \uparrow & & \uparrow s(f_i) \\ U & \xrightarrow{\alpha} & V \end{array} \quad \text{Figure 2}$$

Note that in finite dimensional vector spaces every linear map can be made into a matrix and every matrix can be made into a linear map in this way.

Now suppose U, V, W are finite-dimensional vector spaces over $\mathbb{F}$ with bases $R = (u_1, u_2, \dots, u_r), S = (v_1, v_2, \dots, v_s), T = (w_1, w_2, \dots, w_t)$ respectively. Let $\alpha : U \rightarrow V$ and $\beta : V \rightarrow W$ be linear maps represented by A and B respectively.

By Figure 3 and associativity of matrix multiplication, BA is a matrix representing the linear map $\beta\alpha$.

$$\begin{array}{ccccc} \mathbb{F}^r & \xrightarrow{A} & \mathbb{F}^s & \xrightarrow{B} & \mathbb{F}^t \\ s(R) \uparrow & & \uparrow s(S) & & \uparrow s(T) \\ U & \xrightarrow{\alpha} & V & \xrightarrow{\beta} & W \end{array} \quad \text{Figure 3}$$

See level 7 for discussion and proof of the first isomorphism theorem and the rank-nullity theorem. However, that proof relied on geometric intuition. We would like to prove the rank-nullity theorem over arbitrary fields.

Proposition. If $\alpha : U \rightarrow V$ is a linear map between finite dimensional vector spaces over $\mathbb{F}$ then there exist bases $(e_1, e_2, \dots, e_m)$ for U and $(f_1, f_2, \dots, f_n)$ for V such that α is represented by the block matrix

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

where r is the rank of α .

Note that this immediately implies that $r(\alpha) + n(\alpha) = \dim(U)$.

Proof. Let $e_{k+1}, e_{k+2}, \dots, e_m$ be a basis for $\ker(\alpha)$. Then we can extend this to a basis $e_1, e_2, \dots, e_m$ of U .

Let $f_i = \alpha(e_i)$ for $1 \leq i \leq k$. Then we claim this is a basis for $\text{im}(\alpha)$ as this implies the result. As usual, to show a basis we have to show that it spans and that it is linearly independent.

Spans:

Let $v \in \text{im}(\alpha)$. Then there exist $\lambda_i \in \mathbb{F}$ such that $v = \alpha\left(\sum_{i=1}^k \lambda_i e_i\right)$ since the other basis elements are in the kernel of α . By linearity this implies that $v = \sum \lambda_i \alpha(e_i) = \sum \lambda_i f_i$.

Linear independence:

Suppose that $\sum_{i=1}^k \mu_i f_i = 0$. Then $\alpha\left(\sum_{i=1}^k \mu_i e_i\right) = 0$ so $\sum_{i=1}^k \mu_i e_i \in \ker(\alpha)$. Since $(e_{k+1}, e_{k+2}, \dots, e_m)$ is a basis for $\ker(\alpha)$ we can write

$$\sum_{i=1}^k \mu_i e_i = \sum_{i=k+1}^m \nu_i e_i$$

for some μ_i, ν_i . Since $e_1, e_2, \dots, e_m$ is a basis these must all be 0 so we have linear independence.

Extend $(f_1, f_2, \dots, f_k)$ for a basis of V then we're done.

□

Example. By the Rank-Nullity theorem, the following subspace of $\mathbb{R}^5$ has dimension 3:

$$\{x \in \mathbb{R}^5 : x_1 + x_2 + x_3 = 0 = x_3 - x_4 - x_5\}$$

This is since the map $x \mapsto \begin{pmatrix} x_1 + x_2 + x_3 \\ x_3 - x_4 - x_5 \end{pmatrix}$ has rank 2 so it has a kernel of dimension 3.

Example. If U and W are subspaces of V and V is a finite dimensional subspace over $\mathbb{F}$ and $\oplus$ is taken to be the external direct sum, then the map

$$\begin{aligned} \alpha : U \oplus W &\rightarrow V \\ (u, w) &\mapsto u + w \end{aligned}$$

has image of dimension $\dim(U + W)$ and so its kernel must have dimension $\dim(U \cap W)$.

Now suppose we 2 basis $\{u_i\}$ and $\{e_i\}$ of $\mathbb{F}^m$. Let ι_U be the identity linear map $U \rightarrow U$. Then if P is a change of basis matrix, we can represent this situation by Figure 4:

$$\begin{array}{ccc} U & \xrightarrow{\iota_U} & U \\ \uparrow (\mathbf{u}_i) & & \uparrow (\mathbf{e}_i) \\ \mathbb{F}^m & \xrightarrow{P} & \mathbb{F}^m \end{array}$$

Figure 4

Explicitly, P is a matrix with $u_i = \sum_{k=1}^m P_{ki} e_k$. Take a minute to stare at this diagram and definition until you convince yourself it all works the right way around and then continue.

Now suppose that (e_i) and (f_i) are the standard basis for $\mathbb{F}^m$ and $\mathbb{F}^n$ respectively. Let (u_i) and (v_i) be alternative basis for these spaces respectively and let α be a linear map $U \rightarrow V$ represented by A in the standard basis. Let P be a change of basis matrix with $u_i = \sum_{k=1}^m P_{ki} e_k$ and Q be a change of basis matrix with $v_i = \sum_{k=1}^n Q_{ki} f_k$. Then this situation looks like Figure 5, where B is a matrix representing α with respect to the u and v basis.

$$\begin{array}{ccccccc} U & \xrightarrow{\iota_U} & U & \xrightarrow{\alpha} & V & \xleftarrow{\iota_V} & V \\ \uparrow (\mathbf{u}_i) & & \uparrow (\mathbf{e}_i) & & \uparrow (\mathbf{f}_i) & & \uparrow (\mathbf{v}_i) \\ \mathbb{F}^m & \xrightarrow{P} & \mathbb{F}^m & \xrightarrow{A} & \mathbb{F}^n & \xleftarrow{Q} & \mathbb{F}^n \\ & & & & \searrow B & & \end{array}$$

Figure 5

This implies that $B = Q^{-1}AP$.

Definition. Matrices $A, B \in \text{Mat}_{n,m}(\mathbb{F})$ are said to be equivalent if there exist invertible matrices $P \in \text{Mat}_m(\mathbb{F})$ and $Q \in \text{Mat}_n(\mathbb{F})$ with $B = Q^{-1}AP$.

This means that equivalent matrices can be made to represent the same linear map if you choose the right basis. Also, note that this is an equivalence relation.

Corollary. By this discussion and an earlier proposition, if $A \in \text{Mat}_{n,m}(\mathbb{F})$ then there exist invertible matrices $P \in \text{GL}_m(\mathbb{F})$ and $Q \in \text{GL}_n(\mathbb{F})$ with $Q^{-1}AP = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ for some $0 \leq r \leq \min(m, n)$.

This tells us that there are $\min(m, n)$ orbits of the action of $\text{GL}_m(\mathbb{F}) \times \text{GL}_n(\mathbb{F})$

Corollary. Row rank = Column rank for all matrices

Proof. Let $A \in \text{Mat}_{n,m}(\mathbb{F})$. Then there exist r and invertible P, Q with $Q^{-1}AP = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$. This gives $r = \text{rank}(A)$.

Transposing this equation gives $P^T A^T (Q^T)^{-1} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ which implies that $\text{rank}(A^T) = r$.

□

3 Duality

3.1 Dual space

Definition. (Dual space) Let V be a vector space over a field $\mathbb{F}$. The dual of V is defined as $V^* = \mathcal{L}(V, \mathbb{F}) = \{\theta : V \rightarrow \mathbb{F} : \theta \text{ linear}\}$. Elements of V^* are called linear functionals.

Remark. In normed spaces, we typically define the dual as the set of *continuous* linear maps to $\mathbb{R}$ or $\mathbb{C}$. However, that is a topic for the linear analysis course. In this course we won't worry about continuity.

Example. If $V = \mathbb{R}^3$ and $\theta : V \rightarrow \mathbb{R}$ sends $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \mapsto x_1 - x_3$ then $\theta \in V^*$.

Example. Let $V = \mathbb{F}^X$. Then for any fixed x , θ defined by $\theta(f) = f(x)$ is an element of V^* .

Example. Let $V = C([0, 1])$. Then $f \mapsto \int_0^1 f(t) dt \in V^*$.

Example. The trace $\text{tr} : M_n(\mathbb{F}) \rightarrow \mathbb{F}$ defined by $A \mapsto \sum_{i=1}^n A_{ii}$ is in $M_n(\mathbb{F})^*$.

Proposition. If V is a finite dimensional vector space over $\mathbb{F}$ with basis $(e_1, e_2, \dots, e_n)$ then there is a basis $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$ for V^* such that $\varepsilon_i(e_j) = \delta_{ij}$.

Proof. Since linear maps are characterized by their values on a basis there exist unique choices for $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$. To prove that this is a basis let $\theta \in V^*$, then $\theta = \sum_{i=1}^n \lambda_i \varepsilon_i$ if and only if $\theta(e_j) = \sum_{i=1}^n \lambda_i \varepsilon_i(e_j)$ for all j and this is true if and only if $\lambda_j = \theta(e_j)$. This gives a way to express θ as a linear combination of $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ and proves that this way is unique.

□

Corollary. Finite dimensional vector spaces have the same dimension as their dual.

Consider $\mathbb{F}^n$ where each element is a column vector with respect to the standard basis. Then we can regard elements of

V^* as row vectors $(a_1, a_2, \dots, a_n) = \sum_{j=1}^n a_j \varepsilon_j$ with respect to the dual basis. Then we have that

$$\left(\sum a_j \varepsilon_j \right) \left(\sum x_i e_i \right) = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \sum a_i x_i$$

Proposition. Let V be a finite dimensional vector space over $\mathbb{F}$ with bases $(e_1, e_2, \dots, e_n)$ and $(f_1, f_2, \dots, f_n)$ and that P is a change of basis matrix so that $f_i = \sum_{k=1}^n P_{ki} e_k$. Let $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$ and $(\eta_1, \eta_2, \dots, \eta_n)$ be the corresponding dual basis. Then the change of basis matrix from $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$ to $(\eta_1, \eta_2, \dots, \eta_n)$ is $(P^{-1})^T$ i.e. $\varepsilon_i = \sum_{k=1}^n P_{ik} \eta_k$

Proof. Write $Q = P^{-1}$ so that $e_j = \sum_{k=1}^n Q_{kj} f_k$. Then compute

$$\left(\sum_{k=1}^n P_{ik} \eta_k \right) (e_j) = \left(\sum_{k=1}^n P_{ik} \eta_k \right) \left(\sum_{l=1}^n Q_{lj} f_l \right) = \sum_{k,l} P_{ik} \delta_{lk} Q_{lj} = \sum_{k,l} P_{ik} Q_{lj} = [PQ]_{ij} = \delta_{ij}$$

as required, proving $\varepsilon_i = \sum_{k=1}^n P_{ik} \eta_k$.

□

Definition. (Annihilator) Let V be a vector space and let $U \subseteq V$. Then the annihilator of U is

$$U^0 := \{ \theta \in V^* : \forall u \in U : \theta(u) = 0 \}$$

Note that in $\mathbb{R}^n$ if we consider the standard and dual bases this is isomorphic to the orthogonal complement. If $W \subseteq V^*$ then the annihilator of W is

$$W^0 := \{ v \in V : \forall \theta \in W : \theta(v) = 0 \}$$

Later when we talk about the double dual we will see that the objection that W^0 should be a subset of V^{**} and not V is fine in the case that the vector spaces are finite dimensional.

Example. Consider $\mathbb{R}^3$ with the standard basis e_1, e_2, e_3 and dual basis $\varepsilon_1, \varepsilon_2, \varepsilon_3$. If $U = \langle e_1 + 2e_2 + e_3 \rangle$ and $W = \langle \varepsilon_1 - \varepsilon_3, 2\varepsilon_1 - \varepsilon_2 \rangle$ then $U^0 = W$ and $W^0 = U$.

This example shows the next proposition:

Proposition. Let V be a finite dimensional vector space over $\mathbb{F}$ and U a subspace. Then

$$\dim(U) + \dim(U^0) = \dim(V)$$

Proof. Consider the restriction map $V^* \rightarrow U^*$ given by $\theta \mapsto \theta|_U$. This is linear and since every linear map $U \rightarrow \mathbb{F}$ can be extended to $V \rightarrow \mathbb{F}$ this is a surjection. Moreover the kernel is U_0 so by the rank nullity theorem we have $\dim(V^*) = \dim(U^0) + \dim(U^*)$ and we conclude because $\dim(U^*) = \dim(U)$.

□

3.2 Dual maps

Definition. (Dual map) let V, W be vector spaces over $\mathbb{F}$ and let $\alpha : V \rightarrow W \in \mathcal{L}(V, W)$. Then the dual map to α , written $\alpha^* : W^* \rightarrow V^*$ is given by $\theta \mapsto \theta \circ \alpha$.

It helps to walk through this, since it's hard to grasp the order that everything happens the first time you see it. First, note that θ is in the domain of α^* which is W^* . This means that θ is a map $W \rightarrow \mathbb{F}$. Now α is a map from $V \rightarrow W$ so $\theta \circ \alpha$ is a map $V \rightarrow \mathbb{F}$ hence an element of V^* .

Proposition. Given any θ , $\alpha^*(\theta)$ is a linear map. Also, $\alpha^* \in \mathcal{L}(W^*, V^*)$ is a linear map.

Proof. The proof of the first assertion is left as an exercise.

For the second assertion, note that

$$\alpha^* (\lambda \theta_1 + \mu \theta_2) (v) = (\lambda \theta_1 + \mu \theta_2) (\alpha v) = \lambda \theta_1 \alpha v + \mu \theta_2 \alpha v = (\lambda \alpha^* \theta_1 + \mu \alpha^* \theta_2) v$$

□

Proposition. Let V, W be finite-dimensional vector spaces over $\mathbb{F}$ and $\alpha : V \rightarrow W$ be a linear map. Let $(e_1, e_2, \dots, e_n)$ be a basis for V and $(f_1, f_2, \dots, f_m)$ be a basis for W and let (ε_i) and (η_i) be the corresponding dual bases. Suppose α is represented by A with respect to e_i, f_i , then α^* is represented by A^T with respect to the corresponding dual bases.

Proof. We're given that $\alpha(e_i) = \sum_{k=1}^m A_{ki} f_k$. We want to compute $\alpha^*(\eta_i)$. We have

$$\alpha^*(\eta_i)(e_j) = \eta_i(\alpha(e_j)) = \eta_i\left(\sum_{k=1}^m A_{kj} f_k\right) = \sum_{k=1}^m A_{kj} \delta_{ik} = A_{ij} = \sum_{k=1}^n A_{ik} \varepsilon_k(e_j)$$

Since this is true for all j we have $\alpha^*(\eta_i) = \sum_{k=1}^n A_{ik} \varepsilon_k = \sum_{k=1}^n A_{ki}^T \varepsilon_k$

This implies the result.

□

If $\alpha : U \rightarrow V$ and $\beta : V \rightarrow W$ and $\theta \in W^*$ then

$$(\beta\alpha)^*(\theta) = \theta\beta\alpha = \alpha^*(\theta\beta) = \alpha^*(\beta^*(\theta))$$

This implies that $(\beta\alpha)^* = \alpha^*\beta^*$. This is true for finite dimensions by properties of transposes but we now know it to be true in general vector spaces.

Note that if we change basis i.e. if $B = Q^{-1}AP$ for some invertible P, Q then we have

$$B^T = (Q^{-1}AP)^T = P^T A^T (Q^{-1})^T = \left((P^{-1})^T\right)^{-1} A^T (Q^{-1})^T$$

Therefore the dual change of basis matrices are $(P^{-1})^T$ and $(Q^{-1})^T$.

Proposition. Let $\alpha \in \mathcal{L}(V, W)$ with V, W finite dimensional vector spaces over $\mathbb{F}$. Then $\ker(\alpha^*) = (\text{im}(\alpha))^0$

Proof. If $\theta \in W^*$ then

$$\begin{aligned} \theta \in \ker(\alpha^*) &\iff \alpha^*(\theta) = 0 \iff (\forall v \in V) \theta\alpha(v) = 0 \\ &\iff (\forall w \in \text{im}(\alpha)) \theta(w) = 0 \iff \theta \in (\text{im}(\alpha))^0 \end{aligned}$$

□

Proposition. Let $\alpha \in \mathcal{L}(V, W)$ with V, W finite dimensional vector spaces over $\mathbb{F}$. Then $\text{rank}(\alpha) = \text{rank}(\alpha^*)$

Proof. Follows from “row rank = column rank” property and the dual map = transpose property.

□

Proposition. Let $\alpha \in \mathcal{L}(V, W)$ with V, W finite dimensional vector spaces over $\mathbb{F}$. Then $\ker(\alpha^*) = (\text{im}(\alpha))^0$

Proof. Let $\theta \in \text{im}(\alpha^*)$. Then $\theta = \alpha^*(\phi) = \phi \circ \alpha$ for some $\phi \in W^*$. It helps to think through what is going on here: θ maps V to $\mathbb{F}$ via α which goes to W and ϕ which maps W to $\mathbb{F}$ by the very definition of W^* . This is consistent with θ being in V^* which is the codomain of α^* .

If $v \in \ker(\alpha)$ then

$$\theta(v) = \phi(\alpha(v)) = \phi(0) = 0$$

This implies that $\text{im}(\alpha^*) \subseteq (\ker(\alpha))^0$.

On the other hand,

$$\dim((\ker(\alpha))^0) = \dim(V) - \dim(\ker(\alpha))$$

This implies that

$$\dim((\ker(\alpha))^0) = \dim(\text{im}(\alpha)) = \dim(\text{im}(\alpha^*))$$

which implies that $\ker(\alpha^*) = (\text{im}(\alpha))^0$ because they have the same dimension and one is a subset of the other.

□

3.3 The second dual

Before we talk about taking the dual of the dual, we should discuss what it means for an isomorphism to be canonical or natural. To be honest, I don't really have a good definition, the idea in this case is that it is a natural map between the objects that does not depend on the choice of basis. Basically, although there are isomorphisms between V and V^* there are none that are "special".

We will now define (for finite dimensional vector spaces) a canonical isomorphism $V \rightarrow V^{**}$.

We will define the evaluation map $\text{ev} : V \rightarrow V^{**}$ by $\text{ev}(v)(\theta) = \theta(v)$. To parse this: v is in V and θ is in its dual so that $\theta(v)$ is in $\mathbb{F}$. On the other hand, θ is in V^* and $\text{ev}(v)$ is in V^{**} so we again get something in $\mathbb{F}$ on the left hand side.

Once we can prove that this is actually an isomorphism for finite dimensional vector spaces this will justify our earlier definition of the annihilator of a subset of a dual space.

Note that the evaluation map actually maps V into V^{**} is linear. It is by a straightforward computation that $\text{ev}(v)$ is linear for any fixed v to get that it is in V^{**} and then show that ev itself is linear.

Proposition. If V is finite dimensional then $\text{ev} : V \rightarrow V^{**}$ is an isomorphism.

Proof. Injective:

If $\text{ev}(v) = 0$ for some $v \in V$ then $\theta(v) = \text{ev}(v)(\theta) = 0$ for all $\theta \in V^*$ so $\dim(\langle v \rangle^0) = \dim(V^*) = \dim(V)$ so $\dim(\langle v \rangle) = 0$ so $v = 0$ so ev is injective.

Surjective:

Because V and V^{**} have the same dimension and it is injective.

□

since there is a canonical/basis independent way to choose the isomorphism we will pretend that V and V^{**} are the same with the evaluation map in finite dimensional vector spaces.

Proposition. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Then if $U \leq V$ (this is notation for U is a subspace of V if I haven't said that already) then $U^{00} = U$.

Proof. Let $u \in U$. Then $u(\theta) = \theta(u) = 0$ for all $\theta \in U^0$. So u annihilates everything in U^0 so $u \in U^{00}$ so $U \subseteq U^{00}$. Then an easy check shows that they have the same dimension so they must be equal.

□

Proposition. Let V, W be finite dimensional vector spaces over $\mathbb{F}$. Then if $\alpha \in \mathcal{L}(V, W)$ then $\alpha^{**} = \alpha$ and this is not just because it is the double transpose as we must check that the dual of the dual basis is the original basis.

Proof. Let (e_i) be a basis for V and (f_j) for W with (ε_i) and (η_j) the corresponding dual basis. We have that $e_i(\varepsilon_j) = \delta_{ij} = \varepsilon_j(e_i)$ and $f_i(\eta_j) = \delta_{ij} = \eta_j(f_i)$ so the (e_i) dual is basis to (ε_i) and similar for f, η .

□

Proposition. Let V be a finite dimensional vector space over $\mathbb{F}$ with U_1, U_2 subspaces. Then we have that $(U_1 + U_2)^0 = U_1^0 \cap U_2^0$ and $(U_1 \cap U_2)^0 = U_1^0 + U_2^0$.

Proof. Suppose $\theta \in V^*$. Then

$$\begin{aligned} \theta \in (U_1 + U_2)^0 &\iff (\forall u_1 \in U_1) (\forall u_2 \in U_2) \theta(u_1 + u_2) = 0 \\ &\iff (\forall u \in U_1 \cup U_2) \theta(u) = 0 \iff \theta \in U_1^0 \cap U_2^0 \end{aligned}$$

On the other hand,

$$(U_1 \cap U_2)^0 = (U_1^{00} \cap U_2^{00})^0 = (U_1^0 + U_2^0)^{00} = U_1^0 + U_2^0$$

□

4 Properties of square matrices and endomorphisms

Note: This chapter overlaps a lot with previous work on matrices, all of which is in levels 5.1, 6.1 and 7.3.

4.1 Properties of the determinant

Most of the properties in this course are known from levels 6 and 7.

Proposition. If A is upper triangular then $\det(A) = \prod A_{ii}$

Proof. We know $\det(A) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n A_{i\sigma(i)}$. But $A_{i\sigma(i)} = 0$ whenever $i > \sigma(i)$. But the only permutation for which $i \leq \sigma(i)$ for all i is the identity so that is the only one that contributes to the sum.

□

Definition. A volume form on $\mathbb{F}^n$ is a function $d : (\mathbb{F}^n)^n \rightarrow \mathbb{F}$ that is linear in each of its arguments (i.e. for all i the function $d(v_1, \dots, v_{i-1}, \cdot, v_{i+1}, \dots, v_n)$ is linear, this is called being multilinear) and alternating (i.e. if 2 entries are equal then d gives 0).

From being alternating, we can derive antisymmetry, I'll do it for the $n = 2$ case to illustrate this:

$$\begin{aligned} d(v_1, v_2) &= d(v_1, v_2) + d(v_1, v_1) = d(v_1, v_1 + v_2) = d(v_1, v_1 + v_2) - d(v_1 + v_2, v_1 + v_2) \\ &= d(-v_2, v_1 + v_2) = d(-v_2, v_1 + v_2) + d(-v_2, -v_2) = d(-v_2, v_1) = -d(v_2, v_1) \end{aligned}$$

In the above proof we saw the property that adding a multiple of one argument to another doesn't change the value.

Note that any function that gives the volume scale factor (up to a sign) is a volume form, and therefore must be the determinant. Details are in level 6.1.

Also, see vectors and matrices for a direct proof of $\det(AB) = \det(A) \det(B)$.

We can convert the level 6.1 visual proof of the determinant = signed volume property into an algebra proof as follows:

First, note that for any permutation $\sigma \in S_n$, $d(v_{\sigma(1)}, \dots, v_{\sigma(n)}) = \varepsilon(\sigma) d(v_1, v_2, \dots, v_n)$.

Theorem. Let d be any volume form on $\mathbb{F}^n$ and let $A = (A^{(1)}, A^{(2)}, \dots, A^{(n)}) \in \text{Mat}_n(\mathbb{F})$. Then $d(A) = \det(A) d(I)$ (writing d of a matrix as d of its columns because lazy)

Proof. We compute

$$\begin{aligned} d(A^{(1)}, A^{(2)}, \dots, A^{(n)}) &= d\left(\sum_{i=1}^n A_{i1} e_i, A^{(2)}, \dots, A^{(n)}\right) \\ &= \sum_{i=1}^n A_{i1} d(e_i, A^{(2)}, \dots, A^{(n)}) = \sum_{i,j=1}^n A_{i1} A_{j2} d(e_i, e_j, A^{(3)}, \dots, A^{(n)}) \\ &= \sum_{i_1, i_2, \dots, i_n=1}^n \left(\left(\prod_{j=1}^n A_{i_j j} \right) d(e_{i_1}, e_{i_2}, \dots, e_{i_n}) \right) \end{aligned}$$

We know these are 0 whenever $i_k = i_j$ for some k, j so we are only summing over permutations. This implies that

$$\begin{aligned} &= \sum_{\sigma \in S_n} \left(\left(\prod_{j=1}^n A_{\sigma(j)j} \right) d(e_{\sigma(1)}, e_{\sigma(2)}, \dots, e_{\sigma(n)}) \right) = \sum_{\sigma \in S_n} \left(\left(\prod_{j=1}^n A_{\sigma(j)j} \varepsilon(\sigma) \right) d(e_1, e_2, \dots, e_n) \right) \\ &= \det(A) d(e_1, e_2, \dots, e_n) \end{aligned}$$

as required. □

Now we want to prove that non-trivial kernel implies zero determinant. The reason it's not geometrically obvious is because we are working in arbitrary fields, as with many theorems in this course. From this it will follow that invertible implies non-zero determinant (we know this purely algebraically by the fact that the inverse of a singular matrix has to have $1/0$ determinant) which implies full rank which implies invertible, so they will all be equivalent in all fields. To close this gap, suppose the rank is not full so that the nullity is non-zero so that there is some non-zero vector in the kernel. If $x = (x_1, x_2, \dots, x_n)^T$ satisfies $Ax = 0$ then there is some k such that $x_k \neq 0$. If we define B by $(e_1, e_2, \dots, e_{k-1}, x, e_{k+1}, \dots, e_n)$ then AB has the k 'th column identically 0 so $\det(AB) = 0$ but $\det(AB) = x_k \neq 0$ so $\det(A) = 0$.

Recall the definition of the adjugate matrix from level 6.1 in the proof of the Cayley-Hamilton theorem. It's the result of doing the inverse-finding procedure before dividing by the determinant.

We end with the following useful fact that we kind of know already but it's nice to formalize.

Theorem. Let $A_1, A_2, \dots, A_n$ be blocks, then the determinant of the matrix

$$\begin{pmatrix} A_1 & & \text{stuff} \\ & A_2 & \\ & & \ddots \\ 0 & & & A_n \end{pmatrix}$$

is equal to $\prod_{i=1}^n \det(A_i)$.

Proof. By induction, it suffices to prove that for a block matrix we have $\det \begin{pmatrix} A & C \\ 0 & B \end{pmatrix} = \det(A) \det(B)$.

We could mess around with permutations to prove this, but there is an easier way. We split into 2 cases.

Case 1: If $\det(A) = 0$ then the matrix X does not have full rank so its determinant is 0.

Case 2. If $\det(A) \neq 0$ then the columns of A span all the columns of C so adding multiples of columns of A to the columns with C can make C become 0 without affecting B or the determinant. Therefore,

$$\det(X) = \det \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

To show that the determinant of this is $\det(A)\det(B)$, note that if we take the product over any permutation that does not stay entirely inside A or B it will give 0.

□

4.2 Endomorphisms

Definition. An endomorphism is a linear map $V \rightarrow V$. Here we will study the case where V is finite dimensional and we are working with the same basis. If V is a finite-dimensional vector space over $\mathbb{F}$ then we write $\text{End}(V)$ for the vector space of all linear maps $V \rightarrow V$ and i or I for the identity map.

Definition. Matrices A, B are similar if there is an invertible P with $B = P^{-1}AP$. Note that $\text{GL}_n(\mathbb{F})$ acts on $\text{Mat}_n(\mathbb{F})$ by conjugation and the orbits are the equivalence classes are similarity.

Note that similar matrices have the same characteristic polynomial since the conjugate of λI for any λ is λI so if A and B are similar then $A - \lambda I$ being non-surjective is equivalent to $B - \lambda I$ being non-surjective. Also, this implies that determinant and trace are coordinate-independent.

Proposition. Let $\alpha \in \text{End}(V)$ and $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of α . Then the following are equivalent:

1. α is diagonalizable
2. V has a basis of eigenvectors for α
3. $V = \bigoplus_{i=1}^k E_i$
4. $\dim(V) = \sum_{i=1}^k \dim(E_i)$

Proof. $1 \iff 2$ because a matrix is diagonal exactly when it is being represented with respect to a basis of eigenvectors.

$2 \iff 3$ is clear since eigenspaces are linearly independent as known from earlier levels.

$3 \iff 4$ is by properties of direct sums and linear independence of eigenspaces

□

Note that since a non-zero polynomial can't have more roots than its degree, 2 polynomials in an infinite field are equal if and only if they agree on all roots.

Recall that the minimal polynomial of a matrix is the lowest degree polynomial with leading coefficient 1 such that that polynomial applied to the matrix gives the 0 matrix, and that all such polynomials are multiples of the minimal polynomial and include the characteristic polynomial, and that matrices are diagonalizable if and only if the minimal polynomial has no repeated roots. I proved all this in my vectors and matrices notes.

Now here is a few new theorems that are a bit more exciting.

Theorem. Let $\alpha, \beta \in \text{End}(V)$ be both diagonalizable. Then α and β are simultaneously diagonalizable (i.e. there exists a basis with respect to which both are diagonal) if and only if $\alpha\beta = \beta\alpha$.

Proof. If there exists a basis where α and β are represented by A and B respectively with both diagonal then trivially $AB = BA$ so $\alpha\beta = \beta\alpha$.

On the other hand, suppose $\alpha\beta = \beta\alpha$. Since α is diagonalizable we can decompose it into eigenspaces and so we write

$$V = \bigoplus_{i=1}^k E_{\alpha}(\lambda_i)$$

where λ_i are the eigenvalues of V . Write $E_i := E_{\alpha}(\lambda_i)$. Now for $v \in E_i$ we have $\alpha(\beta(v)) = \beta(\alpha(v)) = \beta(\lambda_i v) = \lambda_i \beta(v)$ which implies that $\beta(E_i) \subseteq E_i$. This implies that $\beta|_{E_i} \in \text{End}(E_i)$. Writing $M_{\beta}(t)$ for the minimal polynomial and $\chi_{\beta}(t)$ for the characteristic polynomial, we have $M_{\beta}(\beta|_{E_i}) = M_{\beta}(\beta)|_{E_i} = 0$. Since $M_{\beta}(t)$ is a product of linear factors we must have that $\beta|_{E_i}$ is diagonalizable, so choose a basis B_i of the eigenvectors for $\beta|_{E_i}$ and do this for all i .

Then since V is a direct sum of the E_i 's we know that $B = \bigcup_{i=1}^k B_i$ is a basis for V consisting of the eigenvectors for both α and β . □

Theorem. An endomorphism α can be represented by an upper triangular matrix if and only if its characteristic polynomial can be written as a product of linear factors (not necessarily distinct). In particular, if $\mathbb{F} = \mathbb{C}$, the fundamental theorem of algebra implies that this is true for every endomorphism.

Proof. Suppose that α is triangulable and represented by

$$\begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Then

$$\chi_{\alpha}(t) = \det \begin{pmatrix} t - \lambda_1 & * & \cdots & * \\ 0 & t - \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & t - \lambda_n \end{pmatrix} = \prod (t - \lambda_i)$$

so it is the product of linear factors.

To prove the converse, we go by induction on n . For $n = 1$ this is trivial.

Suppose $\alpha \in \text{End}(V)$ and the result holds for all spaces of dimensions $< \dim(V)$ and χ_{α} is a product of linear factors. In particular, $\chi_{\alpha}(t)$ has a root, say $\lambda \in \mathbb{F}$. Let $U = E(\lambda) \neq 0$ and let W be a complementary subspace to U in V i.e. $V = U \oplus W$. Let $u_1, u_2, \dots, u_r$ be a basis for U and $w_{r+1}, w_{r+2}, \dots, w_n$ be a basis for W so that $u_1, u_2, \dots, u_r, w_{r+1}, \dots, w_n$ is a basis for V and α is represented by

$$\begin{pmatrix} \lambda I_r & \text{stuff} \\ 0 & B \end{pmatrix}$$

We know $\chi_{\alpha}(t) = (t - \lambda)^r \chi_B(t)$ by properties of the determinant. Therefore $\chi_B(t)$ is a product of linear factors.

Let $\beta : W \rightarrow W$ be represented by B . Then $\dim(W) < \dim(V)$ so by the induction hypothesis there is a basis for W where B is upper triangular. By changing the $w_{r+1}, \dots, w_n$ basis for this basis the result follows. □

Example. The rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is not similar to an upper triangular matrix in $\mathbb{R}$ as this would require it to have real eigenvalues. However, it is similar to $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ in $\mathbb{C}$.

Now recall from vectors and matrices that we can talk about algebraic multiplicity a_{λ} for an eigenvalue which is the number of times it appears as a root of $\chi_{\alpha}(t)$, and we can talk about the geometric multiplicity g_{λ} which is the dimension of the eigenspace and the multiplicity as a root of minimal polynomial which is c_{λ} and we always have $1 \leq c_{\lambda}, g_{\lambda} \leq a_{\lambda}$ with the same proof in arbitrary fields as the one I gave in my vectors and matrices notes.

Before the following example, we recall the classification of Jordan normal forms of 3x3 matrices from Vectors and matrices.

Jordan normal form	Characteristic polynomial	Minimal polynomial
$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$	$(t - \lambda_1)(t - \lambda_2)(t - \lambda_3)$	$(t - \lambda_1)(t - \lambda_2)(t - \lambda_3)$
$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix}$	$(t - \lambda_1)^2(t - \lambda_2)$	$(t - \lambda_1)(t - \lambda_2)$
$\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix}$	$(t - \lambda_1)^2(t - \lambda_2)$	$(t - \lambda_1)^2(t - \lambda_2)$
$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_1 \end{pmatrix}$	$(t - \lambda_1)^3$	$(t - \lambda_1)$
$\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_1 \end{pmatrix}$	$(t - \lambda_1)^3$	$(t - \lambda_1)^2$
$\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 1 \\ 0 & 0 & \lambda_1 \end{pmatrix}$	$(t - \lambda_1)^3$	$(t - \lambda_1)^3$

Recall that for higher size matrices these polynomials might not be enough information.

Example. Consider the matrix over $\mathbb{C}$ given by

$$A = \begin{pmatrix} 3 & -2 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

We want to find its Jordan normal form. Computation gives

$$\chi_A(t) = \begin{vmatrix} t-3 & -2 & 0 \\ 1 & t & 0 \\ 1 & 0 & t-1 \end{vmatrix} = (t-1)((t-3)t+2) = (t-1)^2(t-2)$$

To compute the eigenspaces of A we have $A - I = \begin{pmatrix} 2 & -2 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. This has rank 2 and hence nullity 1, and $(0, 0, 1)^T$ is in the kernel and thus generates it.

For the other eigenspace, we have $A - 2I = \begin{pmatrix} 1 & -2 & 0 \\ 1 & -2 & 0 \\ 1 & 0 & -1 \end{pmatrix}$. This has rank 2 so nullity 1 and for a similar reason its kernel is generated by $(2, 1, 2)^T$. This means that the sum of the dimensions of the eigenspaces is 2, so A is not diagonalizable. This implies that the minimal polynomial of A is $(t-1)^2(t-2)$ since it must have every root of the characteristic polynomial and at least 1 repeated root. This means that its Jordan normal form is

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Recall from vectors and matrices: g_λ is the number of Jordan blocks with eigenvalue λ , a_λ is the sum of their sizes, and c_λ is the size of the largest one. Even for 4×4 matrices, c_λ and a_λ alone is not sufficient because $\begin{pmatrix} 0 & 1 & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}$ and

$\begin{pmatrix} 0 & 1 & & \\ & 0 & & \\ & & 0 & 1 \\ & & & 0 \end{pmatrix}$ have the same minimal and characteristic polynomials but are not similar. If we consider geometric

multiplicity, and exhaustive check of the sizes of Jordan blocks shows that for 6×6 matrices and below we can determine the Jordan normal form uniquely up to reordering of the blocks. But for 7×7 , the matrices

$$\begin{pmatrix} 0 & 1 & & & & \\ & 0 & 1 & & & \\ & & 0 & & & \\ & & & 0 & 1 & \\ & & & & 0 & 1 \\ & & & & & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & 1 & & & & \\ & 0 & 1 & & & \\ & & 0 & & & \\ & & & 0 & 1 & \\ & & & & 0 & 1 \\ & & & & & 0 \end{pmatrix}$$

have the same multiplicity of all 3 types, and yet different Jordan normal forms.

5 Bilinear forms

5.1 Bilinear forms

Definition. (Bilinear form) Let V, W be vector spaces over $\mathbb{F}$. Then a function $\phi : V \times W \rightarrow \mathbb{F}$ is a bilinear form if it is linear in each variable i.e. for each $v \in V$ the map $\phi(v, \cdot) : W \rightarrow \mathbb{F}$ is linear and for each $w \in W$ the map $\phi(\cdot, w) : V \rightarrow \mathbb{F}$ is linear.

Example. The map $V \times V^* \rightarrow \mathbb{F}$ defined by $(v, \theta) = \theta(v)$ is a bilinear form.

Example. The determinant map mapping the columns of a 2×2 matrix to a number is a bilinear form. This is antisymmetric because $\phi(v, w) = -\phi(w, v)$.

Example. Let $V = W = \mathbb{F}^n$. Then the function $(v, w) = \sum_{i=1}^n v_i w_i$ is bilinear.

Example. If $V = W = C([0, 1])$ then $(f, g) \mapsto \int_0^1 fg$ is a bilinear form.

Example. If $A \in \text{Mat}_{m,n}(\mathbb{F})$ then the map $\phi : \mathbb{F}^m \times \mathbb{F}^n \rightarrow \mathbb{F}$ defined by $(v, w) \mapsto v^T A w$ is a bilinear form. Note that the example 2 above this one is a special case of this where $A = I$.

The above example turns out to be all bilinear forms in the following sense.

Definition. Let $(e_1, e_2, \dots, e_n)$ be a basis for V and $(f_1, f_2, \dots, f_m)$ be a basis for W (I need to make a macro write this sentence this is probably at least the 5th time this document bruh) and let $\psi : V \times W \rightarrow \mathbb{F}$ be a bilinear form. Then define a matrix A representing ψ by $A_{ij} = \psi(e_i, f_j)$. Then what happens is that for $v = \sum \lambda_i e_i$ and $w = \sum \mu_j f_j$ we have

$$\psi(v, w) = \psi\left(\sum \lambda_i e_i, w\right) = \sum \lambda_i \psi\left(e_i, \sum \mu_j f_j\right) = \sum_{i,j} \lambda_i \mu_j A_{ij} = \lambda^T A \mu$$

Now we have identified both linear and bilinear maps with matrices, at least in finite dimensional vector spaces. The following proposition is how bilinear maps change basis

Proposition. Let (e_i) and (v_i) be basis for V such that $v_i = \sum_{k=1}^n P_{ki} e_k$ and let (f_j) and (w_j) be basis for W such that $w_j = \sum_{l=1}^m Q_{lj} f_l$. Let $\psi : V \times W \rightarrow \mathbb{F}$ be a bilinear form represented by A with respect to (e_i) and (f_j) and B with respect to (v_i) and (w_j) . Then $B = P^T A Q$.

Proof. We compute

$$\begin{aligned} B_{ij} &= \psi(v_i, w_j) = \psi\left(\sum_{k=1}^n P_{ki} e_k, \sum_{l=1}^m Q_{lj} f_l\right) = \sum_{k,l} P_{ki} Q_{lj} \psi(e_k, f_l) \\ &= \sum_{k,l} P_{ki}^T A_{kl} Q_{lj} = (P^T A Q)_{ij} \end{aligned}$$

□

Note that 2 matrices still represent the same bilinear form with respect to different basis if and only if they are equivalent because if $B = P^{-1}AQ$ then $B = \left((P^{-1})^T\right)^T AQ$.

If we are given a bilinear form $\psi : V \times W \rightarrow \mathbb{F}$ we can define 2 linear maps as follows: Define $\psi_L : V \rightarrow W^*$ by $\psi_L(v) = \psi(v, \cdot)$. Define $\psi_R : W \rightarrow V^*$ by $\psi_R(w) = \psi(\cdot, w)$.

Example. Note that if $\psi : V \times V^* \rightarrow \mathbb{F}$ is defined by $(v, \theta) \mapsto \theta(v)$ then $\psi_L : V \rightarrow V^{**}$ is defined by $v \mapsto \psi(v, \cdot) = \cdot(v)$ which means that ψ_L is the evaluation map. On the other hand, for this ψ , if we consider $\psi_R : V^* \rightarrow V^*$ then we get $\theta \mapsto \psi(\cdot, \theta) = \theta(\cdot)$. This is therefore just the identity map $V^* \rightarrow V^*$.

Proposition. Let (ε_i) be a basis for V^* dual to (e_i) and (η_j) a basis for W^* dual to a basis (f_j) of W . If A represents ψ with respect to (e_i) and (f_j) then A represents ψ_R with respect to (f_j) and (ε_i) and A^T represents ψ_L with respect to (e_i) and (η_j) .

By the way to save me writing in the future, in this document from now on an e, ε basis is taken to have n elements and an f, η basis is taken to have m elements.

Proof. We compute

$$\psi_L(e_i)(f_j) = A_{ij} = \sum_{l=1}^m A_{il}\delta_{lj} = \left(\sum_{l=1}^m A_{il}\eta_l\right)(f_j)$$

This implies that $\psi_L(e_i) = \sum_{l=1}^m A_{il}^T \eta_l$ so A^T represents ψ_L .

We also have

$$\psi_R(f_j)(e_i) = A_{ij} = \sum_{k=1}^n A_{kj}\delta_{ik} = \left(\sum_{k=1}^n A_{kj}\varepsilon_k\right)(e_i)$$

This implies that $\psi_R(f_j) = \sum_{k=1}^n A_{kj}\varepsilon_k$ so A represents ψ_R .

□

Definition. Let ψ be a bilinear form. Then its left kernel is the kernel of ψ_L and its right kernel is the kernel of ψ_R .

More generally, we can write

$$T^\perp = \{w \in W : (\forall t \in T) \psi(t, w) = 0\}$$

$${}^\perp U = \{v \in V : (\forall u \in U) \psi(v, u) = 0\}$$

Then in particular $V^\perp = \ker(\psi_R)$ and ${}^\perp W = \ker(\psi_L)$.

Definition. A bilinear form ψ is said to be non-degenerate if the left and right kernels are both trivial i.e. the 0 space.

Definition. If $\psi : V \times W \rightarrow \mathbb{F}$ is a bilinear form on finite dimensional vector spaces then its rank is the rank of any matrix representing it. This is well defined since $r(P^T A Q) = r(A)$ if P, Q are invertible. This is equivalent to the rank of ψ_L or ψ_R .

Proposition. Let V and W be finite dimensional vector spaces over $\mathbb{F}$ with basis (e_i) and (f_j) respectively. Let $\psi : V \times W \rightarrow \mathbb{F}$ be a bilinear form represented by A with respect to these basis. Then ψ is non-degenerate if and only if A is square and invertible, and this implies V and W have the same dimension.

This basically says if V or W has too many things then some of them are bound to be useless.

Proof. Since ψ_R and ψ_L are represented by A and A^T they both have trivial kernel if and only if A and A^T have 0 nullity, so we need $\dim(V) = r(A) = r(A^T) = \dim(W)$.

□

Example. The 2x2 determinant bilinear form on the reals is non-degenerate because given any non-zero vector we can find another 2D vector to make it non-zero. However, the zero form is degenerate. Also, the dot product is non-degenerate at least in the reals because any vector is non-degenerate if you take its dot product with itself.

5.2 Symmetric bilinear forms and quadratic forms

Definition. Let V be a vector space over $\mathbb{F}$. A bilinear form $\phi : V \times V \rightarrow \mathbb{F}$ is symmetric if $\phi(v, w) = \phi(w, v)$ for all $v, w \in V$.

Example. If S is a symmetric matrix then the bilinear form defined by $\phi(x, y) = x^T S y$ is a symmetric bilinear form.

In fact this example is typical for finite dimensional vector spaces

Proposition. Let V be a finite dimensional vector space over $\mathbb{F}$ and let $\phi : V \times V \rightarrow \mathbb{F}$ be a symmetric bilinear form. Let $(e_1, e_2, \dots, e_n)$ be a basis for V and let M be the matrix representing ϕ with respect to this basis i.e. $M_{ij} = \phi(e_i, e_j)$. Then ϕ is symmetric if and only if M is symmetric.

Proof. If ϕ is symmetric then we compute

$$M_{ij} = \phi(e_i, e_j) = \phi(e_j, e_i) = M_{ji}$$

so M is symmetric.

On the other hand, if M is symmetric then

$$\phi(x, y) = \phi\left(\sum x_i e_i, \sum y_j e_j\right) = \sum_{i,j} x_i M_{ij} y_j = \sum_{i,j} y_j M_{ji} x_i = \phi(y, x)$$

□

Note that by the general change of basis formula proved before, if (e_i) and (f_i) are basis for V such that $f_i = \sum P_{ki} e_k$ and ϕ is represented by A with respect to (e_i) and B with respect to (f_i) then $B = P^T A P$.

Definition. Two square matrices A, B are said to be congruent if there exists some invertible P such that $B = P^T A P$.

Note that this is an equivalence relation and two matrices are congruent if and only if they represent the same bilinear form with respect to different bases. This means that classifying symmetric bilinear forms is the same as classifying symmetric matrices up to congruence.

Definition. A function $q : V \rightarrow \mathbb{F}$ is a quadratic form if there is some bilinear form ϕ such that $q(v) = \phi(v, v)$ for all $v \in V$.

Note that quadratic forms are not linear maps (well, the “zero” one is, but most aren’t).

Example. Let $V = \mathbb{R}^2$ and let ϕ be represented by A with respect to the standard basis. Then

$$q\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = A_{11}x^2 + (A_{12} + A_{21})xy + A_{22}y^2$$

Notice that if we replace A by $\frac{1}{2}(A + A^T)$ then we get a different ϕ but the same q .

Theorem. (Polarization identity) Suppose that $\mathbb{F}$ has the property that $1 + 1 \neq 0$ on $\mathbb{F}$. If $q : V \rightarrow \mathbb{F}$ is a quadratic form then there exists a unique symmetric bilinear form $\phi : V \times V \rightarrow \mathbb{F}$ such that $q(v) = \phi(v, v)$.

Proof. By definition there exists a bilinear form ψ such that $\psi(v, v) = q(v)$. We define $\phi : V \times V \rightarrow \mathbb{F}$ by $\phi(v, w) = \frac{1}{2}(\psi(v, w) + \psi(w, v))$, then this is symmetric and satisfies the condition we want so we have proved the existence part.

Note: It is above where we divide by 2 that we use the fact that $1 + 1 \neq 0$.

To prove uniqueness we can reconstruct ϕ using only q as follows:

$$q(v+w) = \phi(v+w, v+w) = \phi(v, v) + 2\phi(v, w) + \phi(w, w) = q(v) + q(w) + 2\phi(v, w)$$

this means $\phi(v, w)$ can be determined using q and v, w were arbitrary so done.

□

Theorem. Let V be a finite dimensional vector space over $\mathbb{F}$ and suppose $1 + 1 \neq 0$ in $\mathbb{F}$ and let $\phi : V \times V \rightarrow \mathbb{F}$ be a symmetric bilinear form. Then there exists a basis for V such that ϕ is represented by a diagonal matrix with respect to this basis.

Proof. Note that in the reals this follows from the fact that a symmetric matrix is congruent to a diagonal matrix due to being similar by an orthogonal transformation, but this fails over general fields. Therefore, we give this more complicated proof.

Let $n = \dim(V)$. Then we will go by induction on n . For $n = 1$ the theorem is trivial. Suppose this result is true for dimension $< n$.

From the polarization identity we know that if $\phi(v, v) = 0$ for all v then $\phi = 0$. In this case, ϕ will be represented by the 0 matrix with respect to any basis and this is trivially diagonal.

Otherwise, pick $e_1 \in V$ such that $\phi(e_1, e_1) \neq 0$. Define

$$U := \ker(\phi(e_1, \cdot)) = \{u \in V : \phi(e_1, u) = 0\}$$

Since $\phi(e_1, \cdot) \in V^* \setminus \{0\}$ its rank is exactly 1 so we know that $\dim(U) = n - 1$ by the rank-nullity theorem.

Now consider $\phi|_{U \times U} : U \times U \rightarrow \mathbb{F}$ which is a symmetric bilinear form. By the induction hypothesis, there exists a basis $e_2, \dots, e_n$ for U such that $\phi|_{U \times U}$ is represented by a diagonal matrix with respect to this basis. Now by construction, $\phi(e_i, e_j) = 0$ for all $1 \leq i \neq j \leq n$ (if i or j is 1 this is by definition of U) and $(e_1, e_2, \dots, e_n)$ is a basis for V .

□

From now on, we will specialize to $\mathbb{R}$ or $\mathbb{C}$.

Example. Let q be a quadratic form on $\mathbb{R}^3$ given by

$$q\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = x^2 + y^2 + z^2 + 2xy + 4yz + 6xz$$

The reason this is actually a quadratic form is because it is represented by the matrix

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix}$$

To diagonalize this quadratic form we first find f_1 such that $\phi(f_1, f_1) \neq 0$. Direct computation shows that $(1, 0, 0)$ works.

Now we want to find f_2 with $\phi(f_1, f_2) = 0$ and $\phi(f_2, f_2) \neq 0$. This means $(f_2)_1 + (f_2)_2 + 3(f_2)_3 = 0$ because $\phi(f_1, f_2) = 0$. Trying $f_2 = (3, 0, -1)$ we see that this satisfies $\phi(f_2, f_2) = -8$. We have

$$\phi(f_2, v) = \begin{pmatrix} 3 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 3 \\ 1 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = v_2 + 8v_3$$

Finally, we want $\phi(f_1, f_3) = \phi(f_2, f_3) = 0$. This means that $v_2 + 8v_3 = 0$ and $v_1 + v_2 + 3v_3 = 0$. The first condition means we ought to try $f_3 = (x, -8, 1)$, and it turns out with $x = 5$ this satisfies the second condition. Therefore $f_3 = (5, -8, 1)$ works. With these basis elements we have

$$q(af_1 + bf_2 + cf_3) = a^2q(f_1) + b^2q(f_2) + c^2q(f_3) = a^2 - 8b^2 + 8c^2$$

We can also complete the square to get

$$\begin{aligned} x^2 + y^2 + z^2 + 2xy + 4yz + 6xz \\ = (x + y + 3z)^2 - 2yz - 8z^2 \\ = (x + y + 3z)^2 - 8\left(z + \frac{y}{8}\right)^2 + \frac{1}{8}y^2 \end{aligned}$$

We now get the symmetric bilinear form, which we know is the right one by uniqueness:

$$\phi\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}, \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}\right) = (x + y + 3z)(x' + y' + 3z') - 8\left(z + \frac{y}{8}\right)\left(z' + \frac{y'}{8}\right) + \frac{1}{8}yy'$$

We can now use this to find f_1, f_2, f_3 such that $\phi(f_i, f_j) = \delta_{ij}$. If the matrix corresponding to the systems of equations is not invertible, we are in a degenerate case. For f_1 to make $\phi(f_1, f_j) = \delta_{1j}$ we solve

$$\begin{aligned} x + y + 3z &= 1 \\ z + \frac{1}{8}y &= 0 \\ y &= 0 \end{aligned}$$

This gives $f_1 = (1, 0, 0)^T$.

Doing the same for f_2, f_3 gives $f_2 = (-3, 0, 1)^T$ and $f_3 = (-5/8, 1, 1/8)^T$. With respect to this basis we have

$$q(af_1 + bf_2 + cf_3) = a^2 - 8b^2 + \frac{1}{8}c^2$$

From this we see that the diagonal matrix we get is not unique.

Proposition. Let ϕ be a symmetric bilinear form over a complex vector space V . Then there exists a basis $v_1, v_2, \dots, v_m$ for V such that ϕ is represented by

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

with respect to this basis where $r = \text{rank}(\phi)$.

Proof. We already know there exists a basis (e_i) with $\phi(e_i, e_j) = \lambda_i \delta_{ij}$ for some λ_i . By reordering we can assume that $\lambda_1, \lambda_2, \dots, \lambda_r \neq 0$ and $\lambda_{r+1}, \dots, \lambda_n = 0$. For each $1 \leq i \leq r$ pick some complex μ_i with $\mu_i^2 = \lambda_i$ and for $r+1 \leq i \leq n$ let $\mu_i = 1$. Then define $v_i = \frac{e_i}{\mu_i}$. Then we have that $\phi(v_i, v_j) = \frac{1}{\mu_i \mu_j} \phi(e_i, e_j)$ and this is 1 if $i = j \leq r$ and 0 otherwise.

□

Note that for the corresponding quadratic form q we have $q(\sum_{i=1}^n a_i v_i) = \sum_{i=1}^r a_i^2$.

Corollary. In $\mathbb{C}$ every symmetric matrix is congruent to a unique matrix of the form $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$.

Proposition. Let ϕ be a symmetric bilinear form on a finite dimensional vector space over $\mathbb{R}$. Then there exists a basis $(v_1, v_2, \dots, v_n)$ for V such that ϕ is represented by

$$\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

where $p + q = r(\theta)$. The corresponding quadratic form is given by $q(\sum_{i=1}^n a_i v_i) = \sum_{i=1}^p a_i^2 - \sum_{j=p+1}^q a_j^2$.

Proof. We already know there exists a basis (e_i) with $\phi(e_i, e_j) = \lambda_i \delta_{ij}$ for some λ_i . By reordering we can assume that $\lambda_1, \lambda_2, \dots, \lambda_p > 0$ and $\lambda_{p+1}, \dots, \lambda_r < 0$ and $\lambda_{r+1}, \dots, \lambda_n = 0$. Then let μ_i be defined by $\sqrt{\lambda_i}$ for $1 \leq i \leq p$ and $\sqrt{-\lambda_i}$ for $p+1 \leq i \leq r$ and 1 for $i > r$ and define $v_i = \frac{1}{\mu_i} e_i$. Then we find that ϕ is indeed represented by

$$\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

□

Definition. Let ϕ be a symmetric bilinear form on a finite-dimensional real vector space V . We say

1. ϕ is positive semi-definite if $\phi(v, v) \geq 0$ for all $v \in V$
2. ϕ is positive definite if ϕ is positive semi-definite and $\phi(v, v) = 0 \iff v = 0$
3. ϕ is negative semi-definite if $\phi(v, v) \leq 0$ for all $v \in V$
4. ϕ is negative definite if ϕ is negative semi-definite and $\phi(v, v) = 0 \iff v = 0$

Note that if $p = 0$ then we are negative semidefinite. If $p = 0$ and $q = n$ then we are negative definite.

Theorem. (Sylvester's law of inertia) Let ϕ be a symmetric bilinear form on a finite dimensional real vector space V . Then there exists unique non-negative integers p, q such that ϕ is represented by

$$\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with respect to some basis.

Proof. We just need to prove uniqueness. We already know that $p + q = \text{rank}(\phi)$ does not depend on the basis, so it suffices to prove p is unique. To do this, we will characterize p in a basis-independent way by proving that p is the largest dimension of a subspace $P \subseteq V$ such that $\phi|_{P \times P}$ is positive definite.

Note that since ϕ is represented by $\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{pmatrix}$ with respect to some basis (e_i) , the span of $(e_1, e_2, \dots, e_p)$ gives such a P with dimension P , so such a P exists with dimension exactly p . It remains to prove that any such P must satisfy $\dim(P) \leq p$.

To do this, suppose P is any subspace of V such that $\phi|_{P \times P}$ is positive definite. Let $Q = \langle e_{p+1}, e_{p+2}, \dots, e_n \rangle$. Then ϕ restricted to $Q \times Q$ is represented by

$$\begin{pmatrix} -I_q & 0 \\ 0 & 0 \end{pmatrix}$$

Now if $v \in P \cap Q \setminus \{0\}$ then $\phi(v, v) > 0$ since $v \in P \setminus \{0\}$ but $\phi(v, v) \leq 0$ since $v \in Q$ which is a contradiction so this implies $P \cap Q = 0$. This implies that

$$n = \dim(V) \geq \dim(P + Q) = \dim(P) + \dim(Q) = \dim(P) + (n - p)$$

Rearranging gives $\dim(P) \leq p$ as required.

□

Definition. The signature of a bilinear form ϕ is the number $p - q$ where p and q are as above.

Corollary. Every real symmetric matrix is congruent to precisely one matrix of the form

$$\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

5.3 Hermitian forms

Definition. Let V, W be complex vector spaces. Then a sesquilinear form is a function $\phi : V \times W \rightarrow \mathbb{C}$ such that²

1. $\phi(\lambda v_1 + \mu v_2, w) = \bar{\lambda}\phi(v_1, w) + \bar{\mu}\phi(v_2, w)$
2. $\phi(v, \lambda w_1 + \mu w_2) = \lambda\phi(v, w_1) + \mu\phi(v, w_2)$

Definition. Let V, W be finite dimensional complex vector spaces with basis $(v_1, v_2, \dots, v_n)$ and $(w_1, w_2, \dots, w_m)$ respectively and $\phi : V \times W \rightarrow \mathbb{C}$ be a sesquilinear form. Then the matrix representing ϕ with respect to these bases is $A_{ij} = \phi(v_i, w_j)$.

If $v = \sum \lambda_i v_i$ and $w = \sum \mu_j w_j$ then we have

$$\phi(v, w) = \sum_{i,j} \bar{\lambda}_i \mu_j \phi(v_i, w_j) = \lambda^\dagger A \mu$$

Note that symmetric doesn't really make sense anymore as if $\phi(v, w) \neq 0$ then $\phi(iv, w) \neq \phi(v, iw)$. We therefore have this slightly different definition:

Definition. A sesquilinear form on $V \times V$ is Hermitian if $\phi(v, w) = \overline{\phi(w, v)}$.

Note that if ϕ is Hermitian then $\phi(v, v) = \overline{\phi(v, v)} \in \mathbb{R}$ for any $v \in V$. Therefore it makes sense to ask whether it is positive or negative. Also, for any complex number λ we have $\phi(\lambda v, \lambda v) = |\lambda|^2 \phi(v, v)$ so multiplying by a scalar does not change the sign.

We define positive (semi-)definite and negative (semi-)definite Hermitian forms the same way as in the real case.

Proposition. Let $\phi : V \times V \rightarrow \mathbb{C}$ be a sesquilinear form on a finite-dimensional vector space over $\mathbb{C}$ and let (e_i) be a basis for V . Then ϕ is Hermitian if and only if the matrix A representing ϕ is a Hermitian matrix.

Proof. If ϕ is Hermitian then

$$A_{ij} = \phi(e_i, e_j) = \overline{\phi(e_j, e_i)} = A_{ji}^\dagger$$

If A is Hermitian then

$$\phi\left(\sum \lambda_i e_i, \sum \mu_j e_j\right) = \lambda^\dagger A \mu = \overline{\mu^\dagger A^\dagger \lambda} = \overline{\phi\left(\sum \mu_j e_j, \sum \lambda_i e_i\right)}$$

□

Proposition. Let ϕ be a Hermitian form on a finite dimensional vector space V and let (e_i) and (v_i) be bases for V such that $v_i = \sum_{k=1}^n P_{ki} e_k$ and A, B represent ϕ with respect to (e_i) and (v_i) respectively. Then $B = P^\dagger A P$.

Proof. We compute

$$B_{ij} = \phi(v_i, v_j) = \phi\left(\sum P_{ki} e_k, \sum P_{lj} e_l\right) = \sum_{k,l} \overline{P_{ki}} A_{kl} P_{lj} = (P^\dagger A P)_{ij}$$

□

Theorem. (Polarization identity v2) A Hermitian form ϕ on V is determined by the function $\psi : v \mapsto \phi(v, v)$.

Proof. This proof is kind of hard to come up with and fell from the sky but here is the relevant computation.

$$\begin{aligned} \psi(x+y) &= \phi(x, x) + \phi(x, y) + \phi(y, x) + \phi(y, y) \\ -\psi(x-y) &= -\phi(x, x) + \phi(x, y) + \phi(y, x) - \phi(y, y) \end{aligned}$$

²You will see when we get to the linear analysis course that mathematicians are very annoying with this because the convention of which entry is the conjugate linear one is not consistent.

$$\begin{aligned} i\psi(x-iy) &= i\phi(x,x) + \phi(x,y) - \phi(y,x) + i\phi(y,y) \\ -i\psi(x+iy) &= -i\phi(x,x) + \phi(x,y) - \phi(y,x) - i\phi(y,y) \end{aligned}$$

Combining these gives

$$\phi(x,y) = \frac{1}{4}(\psi(x+y) - \psi(x-y) + i\psi(x-iy) - i\psi(x+iy))$$

□

Proposition. (Hermitian form of Sylvester's law of inertia) Let V be a finite dimensional complex vector space and ϕ a Hermitian form on V . Then there exists unique non-negative integers p and q such that ϕ is represented by

$$\begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with respect to some basis.

Proof. The same proof as with symmetric bilinear forms over $\mathbb{R}$.

□

6 Inner product spaces

6.1 Definitions and basic properties

Definition. Let V be a vector space. An inner product on V is a positive-definite symmetric bilinear form or a hermitian form. We will write this as (x,y) instead of $\phi(x,y)$. A vector space with an inner product is an inner product space.

Example. The dot product is an inner product.

Example. The function $(f,g) = \int_0^1 f(t)g(t)dt$ on $C([0,1])$ defines an inner product. In fact, for any $w(t)$ continuous, $(f,g) = \int_0^1 w(t)g(t)f(t)dt$ is an inner product.

If V is an inner product space we define a norm by $\|v\| := \sqrt{(v,v)}$. then $\|v\| = 0$ if and only if $v = 0$. The norm determines the inner product by the polarization identity.

Theorem. (Cauchy-Schwarz inequality but this time unlike in vectors and matrices it includes complex vector spaces as well) Let V be an inner product space and let $v, w \in V$. Then

$$|(v,w)| \leq \|v\| \|w\|$$

Equality occurs if and only if v and w are linearly dependent.

Proof. If $w = 0$ this just says $0 = 0$. Otherwise, since the norm is positive definite, for any λ we have

$$0 \leq (v - \lambda w, v - \lambda w) = (v,v) - \bar{\lambda}(w,v) - \lambda(v,w) + |\lambda|^2(w,w)$$

Now define $\lambda := \frac{(w,v)}{(w,w)}$. Then we get

$$0 \leq (v,v) - \frac{|(w,v)|^2}{(w,w)} - \frac{|(w,v)|^2}{(w,w)} + \frac{|(w,v)|^2}{(w,w)}$$

This therefore implies that $|(w,v)|^2 \leq (v,v)(w,w)$.

For the equality condition, linear dependence $\implies$ equality is left as an exercise. As for the other direction, assume equality holds, so $|(v,w)|^2 = \|v\|^2\|w\|^2$. If $w = 0$, the vectors are trivially linearly dependent. We've already established that:

$$\|v - \lambda w\|^2 = (v,v) - \frac{|(w,v)|^2}{(w,w)}$$

Substituting the equality assumption $|(w, v)|^2 = (v, v)(w, w)$ into this equation gives:

$$\|v - \lambda w\|^2 = (v, v) - \frac{(v, v)(w, w)}{(w, w)} = (v, v) - (v, v) = 0$$

By the positive-definite property of the inner product, a vector has a norm of 0 if and only if it is the zero vector. Therefore:

$$v - \lambda w = 0 \implies v = \lambda w$$

This demonstrates that v can be expressed as a scalar multiple of w , meaning v and w are linearly dependent. □

Corollary. (Triangle inequality) Let V be an inner product space and $v, w \in V$. Then $\|v + w\| \leq \|v\| + \|w\|$.

Proof.

$$\|v + w\|^2 = (v + w, v + w) = (v, v) + (v, w) + (w, v) + (w, w) \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2$$

□

Definition. Let V be an inner product space. Then $v, w \in V$ are orthogonal or perpendicular if $(v, w) = 0$.

Definition. Let V be an inner product space. A set $\{v_i : i \in I\}$ is an orthonormal set if for any $i, j \in I$ we have $(v_i, v_j) = \delta_{ij}$.

Note that an orthonormal set is linearly independent with the same proof as in my vectors and matrices notes.

Definition. Let V be an inner product space. A subset of V is an orthonormal basis if it is an orthonormal set and is a basis.

Note that given an orthonormal basis $(v_1, v_2, \dots, v_n)$ for a finite dimensional inner product space V , then if $v = \sum \lambda_i v_i$ then we can get λ_i by (v_i, v) , since $(v_i, v) = \sum_j \lambda_j (v_i, v_j) = \lambda_i$. This means any $v \in V$ can be written as $\sum (v_i, v) v_i$.

Proposition. (Parseval's identity, finite dimensional case) Let V be a finite dimensional inner product space with an orthonormal basis $(v_i)_{i=1}^n$ and let $v, w \in V$. Then

$$(v, w) = \sum_{i=1}^n \overline{(v_i, v)} (v_i, w)$$

In particular, $\|v\|^2 = \sum_{i=1}^n |(v_i, v)|^2$.

Proof.

$$\begin{aligned} (v, w) &= \left(\sum_{i=1}^n (v_i, v) v_i, \sum_{j=1}^n (v_j, w) v_j \right) = \sum_{i,j=1}^n \overline{(v_i, v)} (v_j, w) (v_i, v_j) \\ &= \sum_{i,j=1}^n \overline{(v_i, v)} (v_j, w) \delta_{ij} = \sum_{i=1}^n \overline{(v_i, v)} (v_i, w) \end{aligned}$$

□

6.2 Gram-Schmidt orthogonalization

Theorem. (Gram-Schmidt process) Let V be an inner product space and let $e_1, e_2, \dots$ be a countable linearly independent set. Then there exists an orthonormal set $v_1, v_2, \dots$ with the property that for all k we have $\langle v_1, v_2, \dots, v_k \rangle = \langle e_1, e_2, \dots, e_k \rangle$.

Proof. For $k = 1$ we just take $v_1 = \frac{e_1}{\|e_1\|}$. Then we go by induction. Suppose we have already found $v_1, v_2, \dots, v_k$ that satisfies the property we want. Then define

$$u_{k+1} = e_{k+1} - \sum_{i=1}^k (v_i, e_{k+1}) v_i$$

For each $j \leq k$ we have

$$\begin{aligned} (v_j, u_{k+1}) &= (v_j, e_{k+1}) - \sum_{i=1}^k (v_i, e_{k+1}) (v_j, v_i) = (v_j, e_{k+1}) - \sum_{i=1}^k (v_i, e_{k+1}) \delta_{ij} \\ &= (v_j, e_{k+1}) - (v_j, e_{k+1}) = 0 \end{aligned}$$

We must prove that u_{k+1} is not 0. We have $\langle v_1, v_2, \dots, v_k, u_{k+1} \rangle = \langle v_1, v_2, \dots, v_k, e_{k+1} \rangle$ as e_{k+1} is a linear combination of the v 's and u_{k+1} . By the induction hypothesis, we also have that these are both equal to $\langle e_1, e_2, \dots, e_k, e_{k+1} \rangle$, and this has dimension $k+1$ since the e_i 's are linearly independent. This means we must have u_{k+1} non-zero or else $\langle v_1, v_2, \dots, v_k \rangle$ would span a space of dimension $k+1$ which is impossible. Therefore, $u_{k+1} \neq 0$ which allows us to divide to get $v_{k+1} = \frac{u_{k+1}}{\|u_{k+1}\|}$. This has the required property, so done by induction. □

Corollary. In every finite-dimensional inner product space every orthonormal set can be extended to an orthonormal basis by extending it to a basis then doing Gram-Schmidt.

Definition. Let V be an inner product space and let V_1 and V_2 be subspaces of V . Then V is the orthogonal internal direct sum of V_1 and V_2 if it is a direct sum and $v \in V_1$ and $u \in V_2$ implies $(v, u) = 0$.

The notation for this is $V = V_1 \perp V_2$.

Definition. If W is a subspace of an inner product space V then the orthogonal complement of W in V is the subspace

$$W^\perp = \{v \in V : (\forall w) (v, w) = 0\}$$

Proposition. Let V be a finite dimensional inner product space and let W be a subspace. Then $V = W \perp W^\perp$.

Proof. We need to prove that we have an internal direct sum and that it is an orthogonal one. This means need to prove that $V = W + W^\perp$ and that $(v_1, v_2) = 0$ for all $v_1 \in W$ and $v_2 \in W^\perp$ (note that the second condition implies $W \cap W^\perp = \{0\}$ so that V indeed becomes a direct sum if we can prove these conditions). The second condition is obvious by the definition of $W^\perp$ so it remains to prove $V = W + W^\perp$.

Let $w_1, w_2, \dots, w_k$ be an orthonormal basis for W and pick $v \in V$. Now let $w = \sum_{i=1}^k (w_i, v) w_i$. Then $w \in W$ and we want to prove $v - w \in W^\perp$. For each j , we can compute

$$(w_j, v - w) = (w_j, v) - (w_j, w) = (w_j, v) - \sum_{i=1}^k (w_i, v) (w_j, w_i) = (w_j, v) - (w_j, v) = 0$$

Therefore for any λ_i we have $(\sum \lambda_j w_j, v - w) = 0$. This means $v - w \in W^\perp$. □

Definition. (Orthogonal external direct sum) Let V_1, V_2 be inner product spaces. The orthogonal external direct sum of V_1 and V_2 is the vector space $V_1 \oplus V_2$ with the inner product defined by $(v_1 + v_2, w_1 + w_2) = (v_1, w_1) + (v_2, w_2)$ with $v_1, w_1 \in V_1$ and $v_2, w_2 \in V_2$.

Proposition. Let V be a finite-dimensional inner product space and let $W \leq V$. Let $(e_i)_{i=1}^k$ be an orthonormal basis of W . Let π be the orthonormal projection of V onto W i.e. $\pi : V \rightarrow W$ is a function that satisfies $\ker(\pi) = W^\perp$ and $\pi|_W = \text{id}$. Then π is given by the formula $\pi(v) = \sum_{i=1}^k (e_i, v) e_i$ and for all $v \in V$ and $w \in W$ we have $\|v - \pi(v)\| \leq \|v - w\|$ with equality if and only if $\pi(v) = w$.

Proof. Note that the inequality condition says $\pi(v)$ is the point on W that is closest to v . This makes sense - see figure 6.

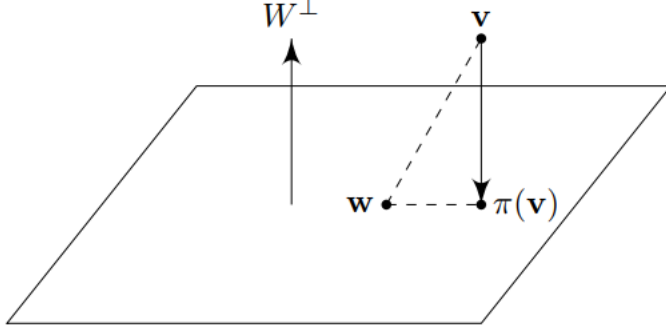


Figure 6

Let $v \in V$ and define $w = \sum_{i=1}^k (e_i, v) e_i$. Then to prove this is $\pi(v)$ we need to prove that $v - w \in W^\perp$. We compute

$$(e_j, v - w) = (e_j, v) - \sum_{i=1}^k (e_i, v) (e_j, e_i) = 0$$

Therefore $v - w$ is orthogonal to every basis vector in W so $v - w \in W^\perp = \ker(\pi)$. This implies

$$\pi(v) = \pi(w) + \pi(v - w) = \pi(w) = w$$

To prove the inequality, note that if x and y are orthogonal then similarly to the triangle inequality proof:

$$\|x + y\|^2 = (x, x) + (x, y) + (y, x) + (y, y) = \|x\|^2 + \|y\|^2$$

This is pythagoras theorem! Now applying this to our projection, for any $w \in W$ and $v \in V$ we have

$$\|v - w\|^2 = \|v - \pi(v)\|^2 + \|\pi(v) - w\|^2 \geq \|v - \pi(v)\|^2$$

These are perpendicular due to the image of π being in W . The equality condition follows from the above display formula.

□

6.3 Adjoints, orthogonal and unitary maps

Proposition. Let V and W be finite-dimensional inner product spaces and let $\alpha : V \rightarrow W$ be a linear map. Then there exists a unique linear map $\alpha^* : W \rightarrow V$ such that

$$(\alpha v, w) = (v, \alpha^* w)$$

for all $v \in V, w \in W$.

Proof. To prove that this is unique if it exists, let $(v_i)_{i=1}^n$ and $(w_j)_{j=1}^m$ be orthonormal basis for V and W respectively and suppose α is represented by A . Suppose $\alpha^* : W \rightarrow V$ satisfies $(\alpha v, w) = (v, \alpha^* w)$ for all $v \in V, w \in W$, then for all i, j , by definition, we know

$$(v_i, \alpha^*(w_j)) = (\alpha(v_i), w_j) = \left(\sum_k A_{ki} w_k, w_j \right) = \sum_k \overline{A_{ki}} (w_k, w_j) = \overline{A_{ji}}$$

This implies that $\alpha^*(w_j) = \sum_i (v_i, \alpha^*(w_j)) v_i = \sum_i \overline{A_{ji}} v_i$. Therefore α^* must be represented by $A^\dagger$ so it is unique.

To show existence we will show that $A^\dagger$ indeed works. We have

$$\begin{aligned} \left(\alpha \left(\sum_i \lambda_i v_i \right), \sum_j \mu_j w_j \right) &= \sum_{i,j} \overline{\lambda_i} \mu_j (\alpha(v_i), w_j) = \sum_{i,j} \overline{\lambda_i} \mu_j \left(\sum_k A_{ki} w_k, w_j \right) = \sum_{i,j} \overline{\lambda_i} \overline{A_{ji}} \mu_j \\ &= \sum_{i,j} \overline{\lambda_i} \mu_j \left(v_i, \sum_k A_{kj}^\dagger v_k \right) = \sum_{i,j} \overline{\lambda_i} \mu_j (v_i, \alpha^*(w_j)) = \left(\sum_i \lambda_i v_i, \alpha^* \left(\sum_j \mu_j w_j \right) \right) \end{aligned}$$

□

Note that the inner product of V defines a natural isomorphism $V \rightarrow \bar{V}^*$ by $v \mapsto (\cdot, v)$

Note that we are defining the complex conjugate of a vector space to have the same elements but such that λv in V is $\bar{\lambda}v$ in $\bar{V}$.

Definition. We call the map α^* the adjoint of α .

Definition. Let V be an inner product space and $\alpha \in \text{End}(V)$. Then α is self-adjoint if $\alpha = \alpha^*$ i.e. $(\alpha(v), w) = (v, \alpha(w))$ for all v, w .

Note that complex matrices are self-adjoint if and only if they are Hermitian and real matrices are self-adjoint if and only if they are symmetric.

Definition. Let V be a real inner product space. Then $\alpha \in \text{End}(V)$ is orthogonal if $(\alpha(v), \alpha(w)) = (v, w)$ for all $v, w \in V$.

By the polarization identity, α is orthogonal if and only if it preserves lengths. A real square matrix is orthogonal if and only if its columns are an orthonormal set.

Proposition. Let V be a finite-dimensional inner product space and let $\alpha \in \text{End}(V)$. Then α is orthogonal if and only if $\alpha^{-1} = \alpha^*$.

Proof. Proved in my groups notes when going through equivalent definitions of orthogonal matrices

□

Note that there is a bijection between orthogonal matrices and orthonormal basis by sending each matrix to the set of its columns.

Definition. Let V be a finite-dimensional complex vector space. Then $\alpha \in \text{End}(V)$ is unitary if $(\alpha(v), \alpha(w)) = (v, w)$ for all $v, w \in V$.

Everything we said above about orthogonal matrices holds for unitary matrices.

6.4 Spectral theory

Proposition. Let V be a finite-dimensional inner product space and let $\alpha \in \text{End}(V)$ be self-adjoint. Then α has a real eigenvalue, all its eigenvalues are real, and eigenvectors with distinct eigenvalues are orthogonal.

Proof. **Step 1.** We prove the eigenvalues are real and one exists if we are working in the complex numbers.

By the fundamental theorem of algebra, α has an eigenvalue, say λ . We pick $v \in V \setminus \{0\}$ such that $\alpha v = \lambda v$. Then

$$\bar{\lambda}(v, v) = (\lambda v, v) = (\alpha v, v) = (v, \alpha v) = (v, \lambda v) = \lambda(v, v)$$

Since $v \neq 0$ we know $(v, v) \neq 0$ so $\lambda = \bar{\lambda}$ therefore λ is real.

Step 2. We prove the eigenvalues are real and one exists if we are working in the real numbers.

Let $e_1, e_2, \dots, e_n$ be an orthonormal basis for V . Then α is represented by a symmetric matrix A with respect to this basis. Since real symmetric matrices are Hermitian viewed as complex matrices, this gives a self-adjoint endomorphism of $\mathbb{C}^n$. By the complex case, A has real eigenvalues only. But the eigenvalues of A are the eigenvalues of α and $M_A(t) = M_\alpha(t)$ since none of these will have complex factors.

Step 2. We prove that eigenvectors with distinct eigenvalues are orthogonal.

Suppose $\alpha v = \lambda v$ and $\alpha w = \mu w$ and $\lambda \neq \mu$. Then we have $(\alpha v, w) = (v, \alpha w)$ by self-adjoint-ness, giving $\lambda(v, w) = \mu(v, w)$ and since $\lambda \neq \mu$ we must have $(v, w) = 0$.

□

Theorem. Let V be a finite-dimensional inner product space and let $\alpha \in \text{End}(V)$ be self-adjoint. Then V has an orthonormal basis of eigenvectors of α .

Proof. Follows from the theorem in my vectors and matrices notes about H being similar to a diagonal matrix by a unitary transformation, and in the real case, it follows from the theorem in my vectors and matrices notes that given a complex eigenvector of a symmetric matrix there exists a real one.

□

Let A be real symmetric. Then since the diagonal values of $P^T A P$ are the eigenvalues of A , the signature of a bilinear form represented by ψ is the number of positive eigenvalues of A minus the number of negative ones.

Proposition. Let V be a finite-dimensional real vector space and let ϕ, ψ be symmetric bilinear forms on V such that ϕ is positive definite. Then there exists a basis $(v_1, v_2, \dots, v_n)$ for V such that both ϕ and ψ are represented by diagonal matrices with respect to this basis.

Proof. We use ϕ to define an inner product. Choose an orthonormal basis $(v_1, v_2, \dots, v_n)$ for V equipped with ϕ with respect to which ψ is diagonal (this is possible because the above theorem implies that given a symmetric matrix A it is congruent to a diagonal matrix D due to being similar by an orthogonal transformation). Then ϕ is represented by I with respect to this basis since $\phi(v_i, v_j) = \delta_{ij}$.

□

Corollary. If A, B are real symmetric matrices and A is positive definite (i.e. has strictly positive eigenvalues) then there exists an invertible matrix Q such that $Q^T A Q$ and $Q^T B Q$ are both diagonal.

Note that the basis is not orthonormal with respect to the standard inner product, but rather the ϕ one, so this Corollary does not mean Q can be chosen to be orthogonal.

Note that by the same argument as the analogous real version of these claims we have

1. If ψ is a Hermitian form on a finite-dimensional complex inner product space V then there is an orthonormal basis for V diagonalizing ψ .
2. If ϕ, ψ are Hermitian forms on a finite-dimensional complex vector space and ϕ is positive definite then there exists a basis where ϕ, ψ are simultaneously diagonalized.
3. If A, B are Hermitian matrices and A is positive definite then there exists invertible Q such that $Q^\dagger A Q$ and $Q^\dagger B Q$ are diagonal.

Theorem. Let V be a finite-dimensional complex vector space and let $\alpha \in \text{End}(V)$ be unitary. Then V has an orthonormal basis of α -eigenvectors.

Proof. By the fundamental theorem of algebra there exists $v \in V \setminus \{0\}$ and $\lambda \in \mathbb{C}$ such that $\alpha v = \lambda v$. By induction assume it holds for dimension $< k$ and as usual note it is trivial for dimension 1.

Now consider $W = \langle v \rangle^\perp$. Then $V = W \perp \langle v \rangle$. We want to show that α restricts to a unitary endomorphism of W . Let $w \in W$. We need to show $\alpha(w)$ is orthogonal to v . We have $(\alpha w, v) = (w, \alpha^{-1} v) = (w, \lambda^{-1} v) = 0$. Therefore, $\alpha(w) \in W$ and $\alpha|_W \in \text{End}(W)$. Also, $\alpha|_W$ is unitary since α is. So by the induction hypothesis, W has an orthonormal basis of α -eigenvectors. If we add $\frac{v}{\|v\|}$ to this basis, we get an orthonormal basis of V with the required property.

□

Note that this fails in the real case since $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ is not real-diagonalizable but is orthogonal.